

A Mathematical Note on `dm_cmb.ipynb`: Patch-Based CMB Prior Learning with Flow Matching

Technical note for the diffusion-model CMB notebook

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Abstract

We document the mathematical content and implementation choices of the notebook `dm_cmb.ipynb`, which learns generative priors for (T, Q, U) patches of the cosmic microwave background (CMB) and Galactic foregrounds at 143 GHz and validates them through flat-sky spectral diagnostics. The workflow combines standard CMB harmonic methods—spin-weighted polarisation decompositions, full-sky simulations from CAMB power spectra, and curved-sky non-Gaussian lensing of the CMB primary by an independently sampled lensing potential through the `lenspyx/DUCC` remapping engine [1–4, 11, 12, 19]—with a modern path-based generative model trained by direct flow matching [8–10]. We describe in detail the pipeline used to bring Planck NPIPE / PR4 half-ring data and simulated priors to a common, well-defined representation: HEALPix pixelisation, harmonic downgrading, beam and pixel-window matching, sky masking, gnomonic patch extraction, spin-2 polarisation rotation, per-channel standardisation, and flat-sky spectrum estimation. The diffusion model is then specified completely: the Gaussian conditional probability path, the choice of signal/noise schedules, the U-Net vector-field architecture, the auxiliary radial Fourier-power regulariser used at training time, the deterministic Heun ODE sampler, and the spectral-validation diagnostics. The eventual scientific goal is posterior sampling for CMB component separation; this note focuses on the prior-learning stage, which is a prerequisite for any likelihood-based inference and which fixes the experimental conventions used in the rest of the work.

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1 Introduction

The cosmic microwave background (CMB) is a relic blackbody radiation field whose temperature and polarisation anisotropies encode the statistical properties of the very early universe and are central to modern cosmology [5, 12]. CMB observations are however contaminated by Galactic foregrounds (most notably thermal dust and synchrotron emission), instrumental noise, and the angular smoothing and sky masking imposed by realistic telescopes. Recovering the underlying CMB signal from the observed maps is therefore a component-separation problem.

In this work we expand on recent applications of deep generative modelling to CMB component separation [18] by exploring the use of *flow-matching* diffusion-style models [8–10] for joint inference of the foreground signal and the true CMB. Let y denote the observed map, x the foreground signal, f the *lensed* CMB, n the instrumental noise, and A the linear instrumental response (effective beam, pixel window, and sky mask). The observed signal model is

$$y = A(f + x) + n, \quad (1)$$

where the gravitational lensing of the CMB is treated implicitly through the prior $p(f)$: the simulated f -prior used for training is constructed by drawing unlensed Gaussian primary CMB realisations together with an independent realisation of the lensing potential ϕ from a fiducial $C_L^{\phi\phi}$ and then remapping the unlensed sky onto the deflected sphere with the `lenspyx` curved-sky lensing engine [19] (Sec. 4.2). The forward operator A therefore remains a linear post-lensing operator (effective beam, pixel window, and sky mask), while the non-Gaussianity induced by lensing on the CMB acoustic peaks, on the $E \rightarrow B$ leakage and on the $T/E \times \phi$ four-point function is carried by the simulated prior. Assuming a Gaussian noise model, the likelihood reads

$$p(y | x, f) \propto \exp\left[-\frac{1}{2} (y - A(f + x))^T N^{-1} (y - A(f + x))\right], \quad (2)$$

with N the noise covariance, and the posterior of interest factorises as

$$p(x, f | y) \propto p(y | x, f) p(x) p(f). \quad (3)$$

Component separation thus requires (i) a tractable likelihood, which is fixed once the instrumental pipeline and noise model are specified, and (ii) data-driven priors $p(x)$ and $p(f)$, which we learn from realistic simulations. We train two independent U-Net models to parametrise vector fields u_t^θ that generate probability paths from a simple Gaussian base distribution to each prior, and we sample those priors by integrating the corresponding ODE/SDE [10]. Posterior sampling is then obtained by augmenting the prior sampler with likelihood gradients $\nabla_x \log p(y | x, f)$ and $\nabla_f \log p(y | x, f)$ during integration; this is the only stage at which the observed CMB data y enters the inference.

We work at a single frequency, 143 GHz, using Planck NPIPE / PR4 half-ring maps as the observed sky y and the publicly released 143 GHz confidence mask as A 's sky cut [5]. Theoretical CMB spectra are computed with CAMB [4] for a fiducial Λ CDM cosmology and used, together with the lensing-potential auto-spectrum $C_L^{\phi\phi}$, to draw lensed-CMB f -priors that share the same effective beam, pixel window, and sky footprint as the observations [11, 19]. Galactic foreground priors are obtained from PySM3 [6, 7], which provides full-sky (I, Q, U) simulations of dust and synchrotron emission consistent with Planck data products; we use the small-scale stochastic presets `d10` and `s5`, so that successive foreground realisations are statistically independent draws of the small-scale Galactic field rather than rotated copies of a fixed template. All maps are processed in the HEALPix pixelisation [3, 13] and the diffusion model is trained on flat-sky 64×64 patches obtained by gnomonic projection.

The scientific motivation is twofold. First, by learning $p(f)$ and $p(x)$ from data we obtain priors that capture non-Gaussian morphology that simple parametric models cannot represent, in particular for Galactic foregrounds. Second, the resulting posterior sampler outputs an ensemble of plausible component maps consistent with the data, providing both a point estimate of (x, f) and a calibrated uncertainty on it. This is a prerequisite for unbiased downstream cosmological inference, including the search for primordial B -modes for which residual foreground contamination is the dominant systematic.

2 Idealized CMB Power Spectra

We briefly review the spherical-harmonic description of CMB temperature and polarisation anisotropies, since this is the natural language in which our priors are constructed and validated. A more pedagogical exposition can be found in Refs. [3, 11, 12].

The CMB temperature anisotropy is a scalar field on the sphere and is therefore naturally expanded in spherical harmonics [1, 3, 12]:

$$T(\hat{n}) = \sum_{\ell=0}^{\infty} \sum_{m=-\ell}^{\ell} a_{\ell m}^T Y_{\ell m}(\hat{n}). \quad (4)$$

From the orthonormality of the spherical-harmonics

$$\int_{\mathbb{S}^2} Y_{\ell m}(\hat{n}) Y_{\ell' m'}^*(\hat{n}) d\Omega = \delta_{\ell\ell'} \delta_{mm'}, \quad (5)$$

the harmonic coefficients are obtained by

$$a_{\ell m}^T = \int_{\mathbb{S}^2} T(\hat{n}) Y_{\ell m}^*(\hat{n}) d\Omega. \quad (6)$$

The coefficient $a_{\ell m}^T$ is therefore the amplitude with which the angular pattern $Y_{\ell m}$ appears in the sky, and in a statistically isotropic theory the spherical-harmonic basis is preferred because two-point statistics become diagonal in (ℓ, m) . The multipole ℓ controls angular scale through the rough correspondence

$$\theta \sim \frac{\pi}{\ell}, \quad (7)$$

so small ℓ corresponds to large coherent angular structures and large ℓ to fine structure; the azimuthal index m resolves the angular dependence within a fixed ℓ .

CMB polarisation is described by the Stokes parameters Q and U defined relative to a chosen local basis on the sphere: Q measures the linear polarisation aligned with the basis (up/down minus left/right), and U the same quantity rotated by 45° . Unlike T , the pair (Q, U) is not a scalar but transforms as a *spin*- ± 2 field under local rotations of the basis, and its natural harmonic expansion uses spin-weighted spherical harmonics [1, 2, 11]:

$$(Q \pm iU)(\hat{n}) = \sum_{\ell m} a_{\ell m}^{\pm 2} Y_{\ell m}(\hat{n}), \quad (8)$$

with coefficients obtained by projection,

$$a_{\ell m}^{\pm 2} = \int_{\mathbb{S}^2} (Q \pm iU)(\hat{n}) {}_{\pm 2}Y_{\ell m}^*(\hat{n}) d\Omega. \quad (9)$$

The scalar E - and pseudo-scalar B -mode coefficients are defined by [1, 2]

$$a_{\ell m}^E = -\frac{1}{2} (a_{\ell m}^2 + a_{\ell m}^{-2}), \quad a_{\ell m}^B = \frac{i}{2} (a_{\ell m}^2 - a_{\ell m}^{-2}). \quad (10)$$

E is the curl-free, parity-even component of the polarisation and B the divergence-free, parity-odd component. The decomposition is local in harmonic space but non-local in pixel space, so finite sky coverage induces E/B mode mixing in any flat-sky or masked estimator.

For a statistically isotropic Gaussian CMB, the harmonic coefficients satisfy

$$\mathbb{E}[a_{\ell m}^X (a_{\ell' m'}^Y)^*] = C_\ell^{XY} \delta_{\ell\ell'} \delta_{mm'}, \quad X, Y \in \{T, E, B\} \quad (11)$$

and all statistical information is contained in the covariance matrix at each ℓ :

$$\Sigma_\ell = \begin{pmatrix} C_\ell^{TT} & C_\ell^{TE} & 0 \\ C_\ell^{TE} & C_\ell^{EE} & 0 \\ 0 & 0 & C_\ell^{BB} \end{pmatrix}. \quad (12)$$

We can compute two-point functions in terms of the C_ℓ^{XY} coefficients. The full temperature

two-point function for instance is:

$$\xi_T(\hat{n}, \hat{n}') := \mathbb{E}[T(\hat{n})T(\hat{n}')] \quad (13)$$

$$\begin{aligned} &= \mathbb{E} \left[\sum_{\ell m} a_{\ell m}^T Y_{\ell m}(\hat{n}) \sum_{\ell' m'} (a_{\ell' m'}^T)^* Y_{\ell' m'}^*(\hat{n}') \right] \\ &= \sum_{\ell m} \sum_{\ell' m'} \mathbb{E}[a_{\ell m}^T (a_{\ell' m'}^T)^*] Y_{\ell m}(\hat{n}) Y_{\ell' m'}^*(\hat{n}') \\ &= \sum_{\ell=0}^{\infty} C_{\ell}^{TT} \sum_{m=-\ell}^{\ell} Y_{\ell m}(\hat{n}) Y_{\ell m}^*(\hat{n}') \\ &= \sum_{\ell=0}^{\infty} \frac{2\ell+1}{4\pi} C_{\ell}^{TT} P_{\ell}(\hat{n} \cdot \hat{n}'). \end{aligned} \quad (14)$$

where we used

$$\sum_{m=-\ell}^{\ell} Y_{\ell m}(\hat{n}) Y_{\ell m}^*(\hat{n}') = \frac{2\ell+1}{4\pi} P_{\ell}(\hat{n} \cdot \hat{n}'). \quad (15)$$

Instead of C_{ℓ}^{XY} it is common to plot

$$D_{\ell}^{XY} \equiv \frac{\ell(\ell+1)}{2\pi} C_{\ell}^{XY}, \quad (16)$$

which is visually more informative because the prefactor $\ell(\ell+1)$ roughly cancels the leading ℓ^{-2} decline of an approximately scale-invariant C_{ℓ} and makes the acoustic peaks easier to see [11].

In the large- ℓ limit, D_{ℓ}^{TT} admits a clean physical interpretation as the temperature variance per logarithmic ℓ -interval. Setting $\hat{n}' = \hat{n}$ in the two-point function and using $P_{\ell}(1) = 1$, we have

$$\text{Var}[T(\hat{n})] = \sum_{\ell=0}^{\infty} \frac{2\ell+1}{4\pi} C_{\ell}^{TT}. \quad (17)$$

For $\ell \gg 1$, approximate the sum by an integral and use $2\ell+1 \simeq 2\ell$:

$$\begin{aligned} \text{Var}[T(\hat{n})] &\simeq \int_0^{\infty} \frac{\ell C_{\ell}^{TT}}{2\pi} d\ell = \int_0^{\infty} \frac{\ell^2 C_{\ell}^{TT}}{2\pi} d \ln \ell \\ &\simeq \int_0^{\infty} \frac{\ell(\ell+1) C_{\ell}^{TT}}{2\pi} d \ln \ell = \int_0^{\infty} D_{\ell}^{TT} d \ln \ell. \end{aligned} \quad (18)$$

The same argument applies to D_{ℓ}^{EE} and D_{ℓ}^{BB} , so the four spectra $\{D_{\ell}^{TT}, D_{\ell}^{EE}, D_{\ell}^{BB}, D_{\ell}^{TE}\}$ provide an interpretable, scale-resolved summary of the second-order statistics of the Gaussian CMB.

Cosmic variance. Even with a perfect noiseless full-sky map, the empirical estimator $\hat{C}_{\ell}^{XY}$ has an irreducible variance [11, 12]

$$\text{Var}[\hat{C}_{\ell}^{XY}] = \frac{1}{2\ell+1} [C_{\ell}^{XX} C_{\ell}^{YY} + (C_{\ell}^{XY})^2], \quad (19)$$

which arises from the finite number $(2\ell+1)$ of independent m -modes available at each multipole. This sets a fundamental floor for the precision with which any model, including the present one, can be tested at a given ℓ , and is largest at low ℓ . On a fraction f_{sky} of the sky, the variance is enlarged by approximately $1/f_{\text{sky}}$, defining the so-called *sample variance* regime relevant to masked analyses [11].

3 Realistic CMB Power Spectra

In practice the idealised description above must be modified to reflect (i) finite-resolution map representations, (ii) instrumental smoothing by the telescope beam, (iii) angular averaging into pixels, (iv) sky cuts that null contaminated regions, and (v) the geometry of patch-based analyses. We summarise here the formalism associated with each of these effects in general terms, independent of any particular data set or simulation; the concrete numerical choices, instrumental parameters, and software implementations adopted in this work are deferred to Sec. 4. The presentation closely follows the conventions of Refs. [3, 11, 12].

3.1 HEALPix pixelisation

We adopt the HEALPix scheme [3, 13] as the all-sky pixelisation throughout. A HEALPix map discretises the sphere into pixels of *equal area* arranged on isolatitude rings, controlled by a single resolution parameter N_{side} (a power of two). The number of pixels and the pixel angular size are

$$N_{\text{pix}} = 12 N_{\text{side}}^2, \quad \theta_r = \sqrt{\frac{4\pi}{N_{\text{pix}}}} = \sqrt{\frac{\pi}{3 N_{\text{side}}^2}}, \quad (20)$$

where θ_r is the side-length of a pixel in radians [3, 11]. Equal-area pixels permit the spherical-harmonic transform to be computed by a single ring-by-ring quadrature without per-pixel weighting, and allow second-order statistics to be averaged uniformly across the sky. HEALPix maps may be stored in either “RING” or “NESTED” ordering; the spherical-harmonic transforms used below assume the RING convention, which is the default for most CMB data products and software [3].

3.2 Band-limit and downgrading

A continuous field on the sphere admits a spherical-harmonic expansion up to $\ell_{\text{max}} = \infty$. A HEALPix map is intrinsically discrete and therefore band-limited; following standard practice one adopts

$$\ell_{\text{max}} = 3 N_{\text{side}} - 1, \quad (21)$$

above which the spherical-harmonic transform is no longer reliable due to aliasing of unresolved structure into the represented modes [3, 11]. Ref. [11] also recommends the more conservative choice $\ell_{\text{max}} = 2 N_{\text{side}}$ when transitioning between resolutions, an inequality that is essentially saturated whenever the beam suppression at the upper end of the represented multipole range is severe.

Harmonic-space downgrading. Bringing a map from a high input resolution $N_{\text{side}}^{\text{in}}$ to a lower target resolution $N_{\text{side}}^{\text{out}}$ is conceptually a band-limit followed by a resampling. The recommended procedure [11] is to perform the downgrade entirely in harmonic space rather than through pixel-space averaging, because pixel-space routines such as `ud_grade` implicitly convolve with the new pixel window and, more importantly, treat Q and U as scalars, breaking the spin-2 transformation law of polarisation and producing biased E/B power. Concretely, one

1. computes polarised harmonic coefficients $(a_{\ell m}^T, a_{\ell m}^E, a_{\ell m}^B)$ of the input map up to

$$\ell_{\text{max}}^{\text{in}} = \min(3 N_{\text{side}}^{\text{in}} - 1, \ell_{\text{max}}^{\text{out}}), \quad (22)$$

i.e. explicitly nulls all multipoles above the target band-limit, as required when synthesising at the lower N_{side} [11];

2. rescales the harmonic coefficients to the desired output beam and pixel window, $a_{\ell m}^{\text{out}} = (B_{\ell}^{\text{out}} P_{\ell}^{\text{out}}) / (B_{\ell}^{\text{in}} P_{\ell}^{\text{in}}) a_{\ell m}^{\text{in}}$ (the Sullivan–Hergt–Scott downgrading recipe [11]), where B_{ℓ} is the beam transfer function of Sec. 3.3 and P_{ℓ} is the HEALPix pixel-window function of Sec. 3.4;
3. synthesises the output (T, Q, U) maps at $N_{\text{side}}^{\text{out}}$ using the rescaled coefficients.

When the input pixel window is sufficiently close to unity in the represented multipole range the second factor reduces to a pure beam rescaling, $a_{\ell m}^{\text{out}} = (B_{\ell}^{\text{out}} / B_{\ell}^{\text{in}}) a_{\ell m}^{\text{in}}$; this approximation should be checked rather than assumed.

3.3 Beam

The optics of any real CMB telescope smear the underlying sky with a finite-width beam, which we model as a circular Gaussian of full-width-half-maximum θ_{FWHM} and Gaussian width σ related by [11]

$$\sigma = \frac{\theta_{\text{FWHM}}}{\sqrt{8 \ln 2}}. \quad (23)$$

For an azimuthally symmetric beam, real-space convolution becomes multiplication in harmonic space, with the scalar transfer function

$$B_{\ell} = \exp\left[-\frac{1}{2} \ell(\ell + 1) \sigma^2\right], \quad (24)$$

so that observed temperature spectra are damped at high ℓ according to

$$C_{\ell}^{TT, \text{obs}} = B_{\ell}^2 C_{\ell}^{TT}. \quad (25)$$

This is the same B_{ℓ} that appears in the harmonic downgrading formula of Sec. 3.2 and in the combined transfer-function model of Sec. 3.4 (Eq. 28). Polarisation, being a spin-2 field, picks up a small additional correction [11, 16]: the polarised beam is approximately ${}_2B_{\ell} = B_{\ell} e^{2\sigma^2}$. At sub-degree resolution this correction is exponentially suppressed and the scalar form (24) is sufficient for both temperature and polarisation; we adopt this approximation in all subsequent expressions.

When data and simulations must be combined coherently, all maps should be brought to a common effective beam. If a map carries an input beam of FWHM θ_{in} and the target resolution is $\theta_{\text{out}} \geq \theta_{\text{in}}$, this is achieved by multiplying its harmonic coefficients by the residual Gaussian

$$B_{\ell}^{\text{add}} = \exp\left[-\frac{1}{2} \ell(\ell + 1) \sigma_{\text{add}}^2\right], \quad \sigma_{\text{add}} = \frac{1}{\sqrt{8 \ln 2}} \sqrt{\theta_{\text{out}}^2 - \theta_{\text{in}}^2}. \quad (26)$$

Enforcing $\theta_{\text{out}} \geq \theta_{\text{in}}$ keeps the operation purely additional smoothing, with no deconvolution and therefore no noise amplification at high ℓ . Bringing the data and the prior simulations to a single, well-defined effective resolution is essential for any likelihood-based component-separation analysis, where the residual $y - A(f + x)$ of the forward model in Sec. 1 must live at a common resolution [11].

3.4 Pixel window

Beam smoothing accounts for the optical blur of the telescope; *pixelisation* introduces an additional, distinct angular smoothing that arises from averaging the sky over the area of each pixel [11]. Working at fixed N_{side} , the corresponding harmonic transfer function $P_{\ell}^{N_{\text{side}}}$ is provided by HEALPix [3] and is well approximated at moderate ℓ by a sinc,

$$P_{\ell}^{\text{sinc}} = \text{sinc}\left(\frac{\ell \theta_r}{2\pi}\right), \quad (27)$$

where θ_r is the pixel size from Eq. (20). The full HEALPix function is computed by spherical-harmonic transform of an average pixel for $N_{\text{side}} \leq 128$ and extrapolated by a top-hat approximation at larger N_{side} [3]. Combining beam and pixel-window contributions, the model for an observed power spectrum becomes

$$C_\ell^{\text{obs}} = P_\ell^2 B_\ell^2 C_\ell^{\text{true}}. \quad (28)$$

Here B_ℓ is the Gaussian beam transfer function of Sec. 3.3 (Eq. 24) and $P_\ell \equiv P_\ell^{N_{\text{side}}}$ is the HEALPix pixel-window function defined above; both operators also appear explicitly in the harmonic downgrading formula of Sec. 3.2. HEALPix [3] provides separate pixel windows P_ℓ^T and P_ℓ^P for scalar and spin-2 fields, and a careful analysis applies the appropriate one to $a_{\ell m}^T$ and to $\{a_{\ell m}^E, a_{\ell m}^B\}$ respectively. The relative magnitudes of B_ℓ and P_ℓ depend on the specific beam and resolution chosen; in regimes where the beam smoothing dominates, the pixel-window correction can be subdominant but is rarely negligible at the highest represented multipoles [11], and should be evaluated quantitatively before being neglected.

3.5 Masking

The Galactic plane and bright extragalactic point sources contaminate the CMB at all frequencies, and any analysis on real data must restrict itself to a sufficiently clean fraction of the sky [5, 11]. This is encoded by a sky mask $M(\hat{n}) \in [0, 1]$, typically a HEALPix map at the resolution of the data, where $M = 1$ flags pixels retained by the analysis and $M < 1$ flags partially or fully contaminated regions. When the mask is provided at higher resolution than the analysis pixelisation, it is downgraded with a scalar pixel-space averaging (`ud_grade`) – legitimate here because M is a scalar weighting and not a spin field – and typically thresholded above some fraction $M_{\text{min}} \in [0, 1]$ to enforce a deterministic clean-sky/contaminated dichotomy.

The choice of how to combine the mask with the power-spectrum estimator is non-trivial. A naive multiplication $\tilde{T} = M T$ followed by a full-sky harmonic transform biases all spectra by approximately a factor of the unmasked sky fraction

$$f_{\text{sky}} = \frac{1}{N_{\text{pix}}} \sum_{i=1}^{N_{\text{pix}}} M_i^2, \quad (29)$$

and additionally couples power across multipoles, leading to ringing and large biases especially for EE/BB where the much weaker polarisation signal sits underneath leakage from the masked-out regions. State-of-the-art alternatives include the pseudo- C_ℓ MASTER estimator [14] as implemented in NaMaster [15], and quasi-maximum-likelihood estimators for the lowest ℓ [11].

A simpler alternative, applicable to patch-based analyses (Sec. 3.6), is to bypass pseudo- C_ℓ altogether by restricting attention to fully-clean patches: candidate patch *centres* are drawn from $\{i : M_i \geq M_{\text{min}}\}$ and any candidate whose projected footprint contains a pixel with $M < M_{\text{min}}$ is rejected. The retained ensemble is then effectively unmasked, and the flat-sky FFT-based spectral estimators of Sec. 3.6 are not biased by mode-coupling. The cost is a non-trivial reduction of usable sky and a small mask-induced bias in the patch ensemble’s prior over sky position, both of which are easy to quantify directly. Apodising the mask before per-patch acceptance and computing pseudo- C_ℓ on each patch with NaMaster [15] would relax the acceptance criterion and recover usable sky in the partially apodised border regions, at the price of restoring a small mode-coupling correction.

3.6 Flat-sky projection and patch spectra

When a generative model or any other translation-invariant learner is trained on small angular regions rather than full-sky maps, the natural data representation is a flat two-dimensional patch. The angular size of a patch is small enough that curvature corrections are subdominant,

while still being large enough to resolve the multipoles of interest, so spherical-harmonic operators can be replaced by translation-invariant 2D Fourier operators at the price of a controlled approximation error.

A patch is obtained by a *gnomonic* (tangent-plane) projection of the underlying HEALPix map onto a square grid of $N_{\text{patch}} \times N_{\text{patch}}$ pixels at angular sampling θ_{patch} , centred on a chosen sky direction $\hat{n}_0$, with linear extent

$$L = N_{\text{patch}} \theta_{\text{patch}}. \quad (30)$$

For $L \lesssim 10^\circ$, curvature corrections to the spherical metric are of order $L^2 \sim \text{few} \times 10^{-2}$ and the flat-sky approximation is quantitatively accurate [11, 17]. The temperature channel is resampled by simple scalar interpolation (e.g. `healpy.projector.GnomonicProj`). For polarisation, however, a naive scalar resampling of Q and U is incorrect: under the change of basis induced by the projection, (Q, U) acquires a position-dependent rotation by an angle $\psi(\hat{n})$, and a *spin-2 rotation*

$$\begin{pmatrix} Q' \\ U' \end{pmatrix} = \begin{pmatrix} \cos 2\psi & \sin 2\psi \\ -\sin 2\psi & \cos 2\psi \end{pmatrix} \begin{pmatrix} Q \\ U \end{pmatrix} \quad (31)$$

must be applied at each pixel [2, 17]. The angle $\psi(\hat{n})$ can be computed analytically or numerically by finite-differencing the projection along the local north direction. Without this rotation, E and B constructed from the patch FFT contain a spurious geometric mixing of the two parities that biases polarised spectra, especially in BB where the signal is intrinsically smaller.

On each patch one estimates spectra via the flat-sky power spectrum estimator [11, 17]: for an angular footprint of side L and pixel size θ_{patch} , the discrete two-dimensional Fourier modes are

$$\ell = 2\pi \mathbf{k}, \quad \mathbf{k} = \left(\frac{n_x}{L}, \frac{n_y}{L} \right) \text{ rad}^{-1}, \quad n_x, n_y \in \mathbb{Z}, \quad (32)$$

and the spin-2 field decomposes into flat-sky E/B via

$$\tilde{E}(\ell) = -\tilde{Q}(\ell) \cos 2\phi_\ell - \tilde{U}(\ell) \sin 2\phi_\ell, \quad \tilde{B}(\ell) = \tilde{Q}(\ell) \sin 2\phi_\ell - \tilde{U}(\ell) \cos 2\phi_\ell, \quad (33)$$

where ϕ_ℓ is the angle of ℓ in the patch frame [2, 17]. The estimator

$$\hat{C}_\ell^{XY} = \frac{\theta_{\text{patch}}^2}{N_{\text{patch}}^2} \langle \tilde{X}(\ell) \tilde{Y}^*(\ell) \rangle_{|\ell| \in \Delta\ell} \quad (34)$$

averages cross-products in annular bins of $|\ell|$ [11], where the angular range is bounded from below by the fundamental mode $\ell_{\min} \simeq 2\pi/L$ and from above by the Nyquist multipole $\ell_{\max} \simeq \pi/\theta_{\text{patch}}$. When patch centres are restricted to clean sky and full-patch validity is enforced (Sec. 3.5), no apodisation or pseudo- C_ℓ correction is required at the patch level; the mode-coupling matrix reduces to the flat-sky identity for an isolated, fully-unmasked patch, and the bias for $f_{\text{sky}}^{\text{patch}} = 1$ is negligible.

3.7 Patch-level standardisation

Independently of the spherical-harmonic formalism, training a generative model requires that the input fields have well-conditioned numerical scales. Raw T , Q , U patches in μK_{CMB} have very different per-channel variances ($\sigma_T \gg \sigma_{Q,U}$ at intermediate frequencies, where the temperature acoustic peaks dominate over the polarised signal), which would force a generative model to learn this trivial scale hierarchy at the cost of higher-order structure and can induce unstable training near schedule endpoints [8, 18]. A standard remedy is the deterministic per-channel standardisation

$$\tilde{X}_c(\mathbf{x}) = \frac{X_c(\mathbf{x}) - \mu_c}{\sigma_c}, \quad c \in \{T, Q, U\}, \quad (35)$$

where μ_c and σ_c are computed once on the training split only and then applied verbatim to any held-out splits. The generator is trained on $\tilde{X}$; at inference time samples are de-normalised by inverting this transformation before computing physical-units diagnostics. The procedure is lossless in the sense that a perfect generator of $p(\tilde{X})$ yields a perfect generator of $p(X)$; it does mean, however, that intermediate spectra plotted in normalised units are scale-rescaled versions of the physical C_ℓ , and the two domains must be distinguished explicitly in any diagnostic.

3.8 Sum–difference noise estimates from independent splits

Modern CMB experiments commonly deliver pairs of independent map-level splits constructed from disjoint subsets of the time-ordered data, e.g. half-mission, half-ring, or odd/even-detector splits. Given two such splits h_1 and h_2 of an observed sky, one forms the standard sum and difference combinations

$$y = \frac{1}{2}(h_1 + h_2), \quad d = \frac{1}{2}(h_1 - h_2). \quad (36)$$

Under the assumption that the sky signal is identical in both halves and the noise is independent between them, y retains the full sky signal and approximately half the noise power, while d is a sky-signal-free realisation of the noise on the map. The map y is therefore the natural observable on which to base inference, while d provides a model-independent diagnostic of the instrumental noise: its empirical variance estimates the diagonal of the per-pixel noise covariance, its auto-spectrum estimates the noise power spectrum N_ℓ , and its lack of cross-correlation with any clean simulation of the sky provides a sanity check on the split assumption.

4 Data

This section instantiates the general formalism of Sec. 3 on the three data products fed to the prior-learning pipeline: the observed Planck 143 GHz sky y , the simulated lensed-CMB prior $p(f)$, and the simulated Galactic-foreground prior $p(x)$. We work at a single observing frequency, 143 GHz, where the 143 GHz channel of Planck has favourable signal-to-noise on the CMB and where the Galactic-foreground contamination is dominated by a narrow set of components (thermal dust and synchrotron) for which standard simulators exist. All maps are stored in HEALPix RING ordering and in μK_{CMB} throughout.

Common processing parameters. By construction all three products live in the same numerical and angular domain (Table 1 summarises the choices). We adopt:

- HEALPix resolution $N_{\text{side}} = 512$, corresponding to $N_{\text{pix}} = 12N_{\text{side}}^2 = 3,145,728$ pixels and a pixel scale $\theta_r \simeq 6.87'$ (Sec. 3.1). This is the lowest power-of-two N_{side} that comfortably band-limits maps smoothed to the target $10'$ beam, while keeping the cost of training a U-Net on patches manageable.
- Harmonic cutoff $\ell_{\text{max}} = 3N_{\text{side}} - 1 = 1535$ in all spherical-harmonic transforms (Sec. 3.2).
- Common effective beam: a circular Gaussian of FWHM $\theta_{\text{out}} = 10'$ applied to all three products via Eq. (24) (Sec. 3.3). This common-beam choice is what makes the data and the prior simulations harmonically compatible.
- Pixel windows: the polarisation-aware HEALPix windows P_ℓ^T and P_ℓ^P at $N_{\text{side}} = 512$ are applied uniformly to both simulated priors $p(f)$ and $p(x)$ via a common harmonic-space step (Sec. 3.4), so that $p(f)$ and $p(x)$ share an identical effective transfer function. At this configuration $P_\ell^{T,P}$ contributes at most an extra $\sim 8\%$ damping at $\ell = \ell_{\text{max}}$ relative to the beam alone. Pixel-window deconvolution is not applied to the observed maps; the corresponding mismatch is bounded by the input pixel-window correction at $N_{\text{side}}^{\text{in}} = 2048$, $\lesssim 0.3\%$ over the represented multipole range.

- Common sky mask: the public Planck 143 GHz confidence mask $M(\hat{n})$, downgraded to $N_{\text{side}} = 512$ with `healpy.ud_grade` and thresholded at $M_{\text{min}} = 0.9$ to define the clean-sky footprint (Sec. 3.5).
- Patch geometry: square $N_{\text{patch}} \times N_{\text{patch}} = 64 \times 64$ flat-sky patches obtained by gnomonic projection at angular sampling $\theta_{\text{patch}} = 6.7'/\text{pixel}$, giving a linear extent

$$L = N_{\text{patch}} \theta_{\text{patch}} \simeq 7.15^\circ \quad \Rightarrow \quad L^2 \simeq 51 \text{ deg}^2, \quad (37)$$

i.e. $f_{\text{sky}}^{\text{patch}} \simeq 0.12\%$ per patch. Curvature corrections are of order $L^2 \sim 1.5 \times 10^{-2}$ and the flat-sky approximation is accurate. Patch projection is implemented by `healpy.projector.GnomonicProj`: the temperature channel is resampled by scalar interpolation, while (Q, U) are first resampled and then spin-2 rotated to the local patch frame using Eq. (34)’s spin-2 matrix, with rotation angle $\psi(\hat{n})$ computed numerically by finite-differencing the projection along the local north direction (function `_project_patch_tqu`, flag `rotate_pol=True`; Sec. 3.6). Non-finite values introduced at projection edges are explicitly replaced by zero.

- Patch-centre selection and full-patch validity: candidate centres are drawn from the set $\{i : M_i \geq 0.9\}$ at $N_{\text{side}} = 512$. For each candidate, the mask is itself first projected onto the same 64×64 gnomonic grid (function `_project_patch_scalar`), and the candidate is rejected unless every pixel of the projected footprint satisfies $M \geq 0.9$ (function `_extract_random_patches`). This produces an ensemble of effectively unmasked patches at the cost of a non-trivial reduction of usable sky, with an empirically observed acceptance rate of order 50–60% at $N_{\text{side}} = 512$ for the 143 GHz mask. A target of $N_{\text{patches}/\text{map}}$ patches is collected per full-sky realisation via rejection sampling, with a hard cap of $40 N_{\text{patches}/\text{map}}$ attempts before raising an error.
- Splits and standardisation: the train/validation split is performed at the level of full-sky realisations rather than at the patch level (a 90/10 split of the N_{maps} realisations, followed by concatenation of the corresponding per-realisation patch banks), so that no large-scale Fourier modes are shared between training and validation; per-channel statistics μ_c, σ_c ($c \in \{T, Q, U\}$) are computed once on the training split only and applied verbatim to the validation split (Sec. 3.7, 4.4). To guard against degenerate channels, σ_c is floored at 10^{-6} before inversion.
- Bandpower binning: when validating a learned prior with the flat-sky estimator (34), we bin into ~ 30 linear or log-spaced annuli of $|\ell|$ between $\ell_{\text{min}} \simeq 2\pi/L \approx 50$ and $\ell_{\text{max}} \simeq \pi/\theta_{\text{patch}} \approx 1500$.

The remainder of this section documents the product-specific processing on top of these common conventions: the Planck half-ring preprocessing (Sec. 4.1), the CAMB-based CMB simulation (Sec. 4.2), and the PySM-based foreground simulation (Sec. 4.3).

4.1 Observed CMB data

We use the publicly released Planck PR4/NPIPE [5] half-ring maps at 143 GHz, h_1 and h_2 , and the corresponding 143 GHz confidence mask, all distributed as HEALPix RING-ordered FITS files at $N_{\text{side}}^{\text{in}} = 2048$. Each half-ring file contains the three Stokes parameters (I, Q, U) in K_{CMB} .

The two half-ring maps are processed independently by the routine `preprocess_pr4_hr_to_nside512`, which implements the harmonic-space “smooth-and-downgrade” procedure of Sec. 3.2:

1. read (I, Q, U) , replace `HEALPix UNSEEN` sentinels and non-finite values by zero, and rescale by 10^6 to convert $\text{K}_{\text{CMB}} \rightarrow \mu\text{K}_{\text{CMB}}$ (units inferred from the FITS TUNIT/BUNIT header or, as a fallback, from the map RMS);
2. compute polarised harmonic coefficients $(a_{\ell m}^T, a_{\ell m}^Q, a_{\ell m}^U)$ with `healpy.map2alm` (`pol=True`, `iter=3`) up to $\ell_{\text{max}} = 3N_{\text{side}} - 1 = 1535$;

Parameter	Value
HEALPix resolution N_{side}	512
Harmonic cutoff ℓ_{max}	1535
Common effective beam θ_{FWHM}	10' Gaussian
Frequency	143 GHz
Sky mask	Planck 143 GHz confidence, $M \geq 0.9$
Patch grid	64×64
Patch sampling θ_{patch}	6.7'/pixel
Patch extent L	$\simeq 7.15^\circ$ ($\simeq 51 \text{ deg}^2$)
Train/val split	90/10
Bandpower bins	~ 30 , $\ell \in [50, 1500]$
Units	μK_{CMB}

Table 1: Summary of common processing parameters used uniformly for the observed sky y , the simulated CMB prior $p(f)$ (Sec. 4.2), and the simulated foreground prior $p(x)$ (Sec. 4.3).

- multiply each coefficient by the residual Gaussian transfer function

$$B_\ell^{\text{add}} = \exp\left[-\frac{1}{2}\ell(\ell+1)\sigma_{\text{add}}^2\right], \quad \sigma_{\text{add}} = \frac{1}{\sqrt{8\ln 2}}\sqrt{\theta_{\text{out}}^2 - \theta_{\text{in}}^2}, \quad (38)$$

with $\theta_{\text{in}} = 7.27'$ and $\theta_{\text{out}} = 10'$ as in Sec. 3.3;

- synthesise (T, Q, U) at $N_{\text{side}} = 512$ via `healpy.alm2map (pol=True)`.

Pixel-window deconvolution is omitted at this stage, an approximation justified in Sec. 4: the input pixel-window correction at $N_{\text{side}}^{\text{in}} = 2048$ is $\lesssim 0.3\%$ throughout the represented multipole range.

From the two preprocessed half-ring maps we form the half-mission-style sum and difference of Sec. 3.8,

$$y = \frac{1}{2}(h_1 + h_2), \quad d = \frac{1}{2}(h_1 - h_2), \quad (39)$$

and use y as the observed sky entering the likelihood of Sec. 1 and d as a sky-signal-free realisation of the instrumental noise. On the clean-sky footprint defined in Sec. 4 we verify the expected coherence properties: h_1 and h_2 are positively correlated pixel-by-pixel in all three Stokes channels; the half-ring difference satisfies $\text{RMS}(d) \ll \text{RMS}(y)$, with the noise-to-signal RMS ratio of order 10% in T and order 1 in Q, U at the 10' resolution; and the binned cross-spectrum $\hat{C}_\ell^{h_1 \times h_2}$ tracks the fiducial CMB spectrum (Sec. 4.2) across $30 \leq \ell \leq \ell_{\text{max}}$ in all four diagonal channels $TT/EE/BB/TE$. These checks fix the data-domain conventions used identically for the two simulated priors below.

4.2 $p(f)$: simulated CMB priors

The CMB prior $p(f)$ is constructed at the level of *maps* that are physically lensed by an independently sampled realisation of the lensing potential ϕ , rather than by drawing Gaussian fields with the lensed angular power spectrum directly. This sub-section spells out the simulation pipeline, motivates the choice over the simpler Gaussian-from-lensed- C_ℓ alternative (which is retained as a legacy/ablation path), and discusses the path towards a fully self-consistent treatment of lensing as an explicit forward-model operator.

Cosmology and CAMB call. We adopt a fiducial flat ΛCDM cosmology with $H_0 = 67.5 \text{ km s}^{-1} \text{ Mpc}^{-1}$, $\Omega_b h^2 = 0.022$, $\Omega_c h^2 = 0.122$, $\sum m_\nu = 0.06 \text{ eV}$, $\tau = 0.06$, $A_s = 2.1 \times 10^{-9}$, $n_s = 0.965$, $r = 0$. A single call to the Boltzmann solver CAMB [4] with `lens_potential_accuracy=1` provides, in μK^2 and on the integer grid $L = 0, \dots, \ell_{\text{max}}$:

- the unlensed scalar CMB spectra $\{C_\ell^{TT,u}, C_\ell^{EE,u}, C_\ell^{BB,u}, C_\ell^{TE,u}\}$ via `get_cmb_power_spectra(..., raw_cl=True)['unlensed_scalar']`; with $r = 0$ the unlensed $C_\ell^{BB,u}$ is identically zero;
- the lensing-potential auto-spectrum $C_L^{\phi\phi}$ via `get_lens_potential_cls(..., raw_cl=True)[: , 0]`;
- for diagnostic purposes only, the lensed total spectra $\{C_\ell^{TT}, C_\ell^{EE}, C_\ell^{BB}, C_\ell^{TE}\}$, which provide the analytic 2-point target that the lensed sims must match in expectation.

The unlensed spectra plus $C_L^{\phi\phi}$ are the only inputs to the lensing pipeline below; the lensed CMB spectra are retained as a validation target only.

Lensed-map sampler. We use the curved-sky CMB-lensing remapping engine `lenspyx` [19], which wraps the high-performance non-uniform spherical-harmonic transforms of `DUCC` and computes the deflected sky to a user-controlled accuracy with cost essentially equal to two standard pol-aware SHTs. Each independent map realisation in the prior bank is generated by the routine `simulate_lensed_cmb_tqu_lenspyx`, which proceeds in five harmonic-space steps:

1. *Bandlimits.* Set the lensing-remap bandlimit to $\ell_{\max} = 3N_{\text{side}} - 1 = 1535$ and zero the unphysical $\ell = 0, 1$ entries of $C_\ell^{TT,u}, C_\ell^{EE,u}, C_\ell^{BB,u}, C_\ell^{TE,u}, C_L^{\phi\phi}$ before any sampling.
2. *Unlensed primary draw.* Draw a single correlated polarised realisation of the unlensed CMB,

$$(a_{\ell m}^{T,u}, a_{\ell m}^{E,u}, a_{\ell m}^{B,u}) \sim \mathcal{N}\left(0, \begin{pmatrix} C_\ell^{TT,u} & C_\ell^{TE,u} & 0 \\ C_\ell^{TE,u} & C_\ell^{EE,u} & 0 \\ 0 & 0 & C_\ell^{BB,u} \end{pmatrix}\right), \quad (40)$$

with `healpy.synalm(new=True)`, exactly as in the legacy Gaussian sampler but with the *unlensed* CMB spectra.

3. *Lensing-potential draw.* Independently draw a Gaussian realisation of the lensing potential,

$$\phi_{LM} \sim \mathcal{N}(0, C_L^{\phi\phi}), \quad (41)$$

by a separate `healpy.synalm` call. The CMB $\times\phi$ cross-spectra $C_L^{\phi T}$ and $C_L^{\phi E}$ are not imposed here, an approximation discussed below.

4. *Deflection field.* The gravitational deflection vector is the gradient of ϕ , $\alpha(\hat{n}) = \nabla\phi(\hat{n})$, with curl-free gradient-mode alm $d_{LM}^{\text{grad}} = \sqrt{L(L+1)}\phi_{LM}$ (curl-mode $\Omega_{LM} = 0$ for scalar lensing). The factor $\sqrt{L(L+1)}$ is precisely the `lenspyx` convention for the input deflection alm and is applied via `healpy.almxfl`.
5. *Remap onto the deflected sphere.* A single call

$$\text{lenspyx.alm2lenmap}([a^{T,u}, a^{E,u}, a^{B,u}], [d^{\text{grad}}])$$

with `geometry=('healpix', {nside: 512}), pol=True` and `epsilon=10-6` produces a *truly lensed* ($T_{\text{lens}}, Q_{\text{lens}}, U_{\text{lens}}$) realisation: the remap is implemented by `ducc0` as a pair of non-uniform SHTs that evaluate the unlensed CMB at the deflected positions and rotate (Q, U) by the appropriate spin-2 phase along the deflection direction.

The output of step (5) is a HEALPix map at $N_{\text{side}} = 512$ that carries the full lensing imprint of the sampled ϕ , not just its diagonal power-spectrum signature. By construction it inherits, in expectation, the lensed CMB power spectra of step (1), but at the level of an individual realisation it also contains all the lensing-induced non-Gaussian structure: the connected 4-point function (trispectrum) that underpins quadratic CMB-lensing reconstruction, the non-Gaussian texture of lensing B -modes generated by $E \rightarrow B$ mode coupling, and the lensing modulation of the small-scale acoustic structure (Sec. 3).

Post-lensing instrumental response. The remap is performed at the native unbeamed, unwindowed resolution. The common observational response is then applied to the lensed map in harmonic space, matching the convention used for y in Sec. 4.1:

1. compute $(a_{\ell m}^T, a_{\ell m}^E, a_{\ell m}^B)$ of $(T_{\text{lens}}, Q_{\text{lens}}, U_{\text{lens}})$ via `healpy.map2alm (pol=True, iter=3)`;
2. multiply by the joint beam-and-pixel-window transfer functions

$$a_{\ell m}^T \rightarrow B_\ell P_\ell^T a_{\ell m}^T, \quad a_{\ell m}^{E,B} \rightarrow B_\ell P_\ell^P a_{\ell m}^{E,B}, \quad (42)$$

with B_ℓ from Eq. (24) at $\theta_{\text{FWHM}} = 10'$ and $P_\ell^{T,P}$ from `healpy.pixwin(., pol=True)`;

3. synthesise the final (T, Q, U) at $N_{\text{side}} = 512$ via `healpy.alm2map (pol=True)`.

This ordering—lens *first*, beam and pixel-window *after*—is the physically correct one: the beam and the HEALPix pixel window act on the observed (and therefore lensed) sky, not on the underlying primary CMB. Each realisation is generated from an independent random seed that drives both the unlensed $(a^{T,u}, a^{E,u}, a^{B,u})$ draw and the independent ϕ draw, so N_f runs yield a statistically independent ensemble at fixed cosmology.

Why not just Gaussian draws from the lensed C_ℓ ? The legacy sampler `simulate_cmb_tqu_from_cls`, which draws Gaussian alm from the *lensed* CAMB spectra and applies beam/pixel-window in one shot, is retained in the codebase for back-compatibility and as a controlled ablation. It is however strictly mis-specified once any non-2-point statistic is exposed to inference:

- It reproduces $\langle C_\ell^{TT/EE/BB/TE} \rangle$ by construction but has identically Gaussian one-point and higher-order n -point statistics, with vanishing lensing trispectrum and vanishing T, E, ϕ -bispectrum at fixed ϕ .
- The lensing-induced B -modes generated by $E \rightarrow B$ leakage have, in real lensed skies, a characteristic non-Gaussian morphology that traces the underlying ϕ field; a Gaussian draw from the same C_ℓ^{BB} produces B -modes with no such structure and is observationally indistinguishable from primordial r -driven B at the 2-point level only.
- Cosmology-marginalisation extensions (varying Ω_m, Ω_Λ , etc.) shift both C_ℓ^{unl} and $C_L^{\phi\phi}$ *consistently*, so the lenspyx pipeline correctly propagates parameter variations to the lensed CMB; the Gaussian-lensed- C_ℓ pipeline only propagates the marginal C_ℓ^{lensed} , losing the correlated parameter dependence of the lensing kernel.

The remapping cost is dominated by two pol-aware SHTs at $N_{\text{side}} = 512$ (a fraction of a second per realisation on a single core through DUCC), so the upgrade is essentially free relative to map I/O and patch extraction. Empirically (notebook validation cell), individual lenspyx realisations recover the lensed CAMB $\{C_\ell^{TT}, C_\ell^{EE}\}$ to within cosmic variance on the full sky; the lensing-induced C_ℓ^{BB} converges from below as the lensing bandwidth approaches the support of $C_L^{\phi\phi} \otimes C_\ell^{EE}$, an artefact of the finite- ℓ_{max} remap that vanishes well before the noise level of Planck 143 GHz.

Approximations and forward look: explicit lensing operator (Option 2). The current pipeline carries one explicit approximation: the unlensed CMB $(a^{T,u}, a^{E,u}, a^{B,u})$ and the lensing potential ϕ_{LM} are drawn *independently*, so the small ($\mathcal{O}(10^{-3})$) CMB $\times\phi$ cross-spectra $C_L^{\phi T}$ (from late-time ISW) and $C_L^{\phi E}$ are not imposed at the level of individual realisations. This is the standard convention used by Planck FFP10 [5] and CMB-S4 simulation libraries, and is negligible compared with cosmic variance at the multipole range used here ($\ell \lesssim 1500$). A subdominant second approximation is that ϕ is treated as Gaussian; the leading-order ϕ non-Gaussianity from non-linear large-scale structure is at the few-percent level on $C_L^{\phi\phi}$ [20] and well below the simulation accuracy required for the present problem.

A more principled treatment, planned as a follow-up extension when polarisation B -modes (and hence the tensor-to-scalar ratio r) become the science target, is to absorb the lensing remap into the forward model rather than into the prior. Concretely, Eq. (1) is rewritten as

$$y = A [\mathcal{L}(\phi) f_u + x] + n, \quad (43)$$

where f_u is the *unlensed* CMB and $\mathcal{L}(\phi)$ is the gravitational-lensing operator built from the lensing potential ϕ (the same operator that `lenspyx.alm2lenmap` implements). The advantages of this “Option 2” are that

1. the prior $p(f_u)$ becomes a Gaussian on the unlensed CMB, which is the physically correct prior and whose density is closed-form—no need to train a neural prior to learn lensing non-Gaussianity, since it is now in the model;
2. the lensing potential ϕ becomes a latent random field with a Gaussian prior $p(\phi) = \mathcal{N}(0, C_L^{\phi\phi})$ that can be jointly inferred from the data, yielding a calibrated posterior $p(f_u, x, \phi | y)$ rather than a marginal $p(f, x | y)$;
3. cosmological-parameter marginalisation is straightforward because all dependence on Ω_m , Ω_Λ , H_0 is now carried by the closed-form Gaussian priors on f_u and ϕ rather than by an empirically learned non-Gaussian prior.

Three sub-variants of Option 2 are natural and would re-use most of the present infrastructure:

- *Fixed- ϕ .* Replace ϕ in Eq. (43) with the public Planck PR4 lensing reconstruction `COM_Lensing_4096_R4.00` [21]. The likelihood gradients used during posterior sampling become $\nabla_{f_u} \log p(y | f_u, x)$ computed through the linear adjoint of $\mathcal{L}(\hat{\phi})$, which `lenspyx` also exposes. Cheapest variant but ignores reconstruction noise on $\hat{\phi}$.
- *Joint ϕ sampling.* Augment the latent state of the diffusion sampler with ϕ and add likelihood gradients $\nabla_\phi \log p(y | f_u, x, \phi)$ alongside the existing $\nabla_{f_u, x} \log p(y | f_u, x, \phi)$. This realises the MUSE / Anderes–Millea–Seljak scheme [22] in the diffusion-sampler framework and yields a fully calibrated joint posterior on (f_u, x, ϕ) .
- *Quadratic-marginalised ϕ .* Marginalise ϕ to leading order analytically (Hu–Okamoto quadratic estimator). Useful as a fast consistency check; the residual non-Gaussianity in f_u after marginalisation is at the level of higher-order ϕ moments and is negligible at Planck noise levels.

The Option-1 sampler implemented in this work is the strict subset of the fixed- ϕ variant in which one further drops the dependence on $\hat{\phi}$ at sampling time and absorbs the resulting marginal lensed CMB into a learned prior. It is the right starting point for the present temperature-dominated analysis, and provides a precise quantitative baseline against which any future Option-2 extension will be validated.

Output. A bank of N_f independent full-sky `lenspyx`-lensed realisations is generated by the routine `build_prior_patch_datasets` (with `lensing_method='lenspyx_remap'`, the default) and reduced to flat (T, Q, U) patches by the common patch-extraction and standardisation pipeline of Sec. 4. The resulting training-ready array has shape $(N_{\text{patches}}, 3, 64, 64)$ with $N_{\text{patches}} = N_f \times N_{\text{patches}/\text{map}}$. Selecting `lensing_method='gaussian_lensed_cl'` switches back to the legacy Gaussian-from-lensed- C_ℓ sampler with the same output schema, useful as an ablation control.

4.3 $p(x)$: simulated foreground priors

The Galactic-foreground prior is built from PySM3 [6, 7], instantiated at $N_{\text{side}} = 512$ with the small-scale component model **d10+s5**: **d10** is a modified-blackbody thermal-dust template whose dust temperature $T_d(\hat{n})$ and spectral index $\beta_d(\hat{n})$ are spatially varying maps derived from Planck data, and whose small-scale fluctuations below the resolution of the input templates ($\ell \gtrsim 300$) are populated stochastically by a non-Gaussian small-scale model fitted to high-latitude Planck residuals; **s5** is the analogous spatially-varying-spectral-index synchrotron model whose small-scale modes are drawn stochastically from a fitted power-law amplitude. The choice of small-scale presets, rather than the deterministic **d1+s1** pair, is critical: it makes each foreground realisation a genuinely independent draw of the small-scale Galactic field, rather than a rotated copy of a single deterministic template, and is what allows the learned prior $p(x)$ to capture the patch-level variability of the Galactic emission. We exclude PySM’s **c1** (lensed-CMB) component from the foreground sky used in $p(x)$, since the lensed CMB is already drawn independently as $p(f)$ and including it twice would bias the joint prior; an exploratory **d1+s1+c1** sky is instantiated at the top of the data section purely as a visual sanity check (Mollweide views of I and $P = \sqrt{Q^2 + U^2}$ at 143 GHz) and is not used downstream.

Independent stochastic realisations. For each of N_x realisations we re-seed the random number generator that drives PySM’s small-scale models with a fresh seed derived deterministically from a master foreground seed, and then build a new **Sky** object. Because PySM samples its small-scale modes at **Sky**-construction time, this guarantees that successive realisations have *statistically independent* small-scale Galactic structure while sharing the same large-scale template above the small-scale-model crossover multipole. In particular, both the morphology of the small-scale dust filaments and the small-scale polarisation amplitude vary realisation-to-realisation, in contrast with the deterministic **d1+s1** presets where these would be repeated identically. Emission is evaluated at the common observing frequency 143 GHz and converted from PySM’s native Rayleigh–Jeans antenna units to thermodynamic μK_{CMB} through the Planck blackbody equivalency, ensuring unit consistency with y and f .

Random orientation augmentation. Each stochastic realisation is then optionally rotated by a uniformly sampled element $R \in \text{SO}(3)$, drawn via Euler angles

$$\alpha \sim \mathcal{U}(0, 2\pi), \quad \cos \beta \sim \mathcal{U}(-1, 1), \quad \gamma \sim \mathcal{U}(0, 2\pi), \quad (44)$$

acting on the full (T, Q, U) stack with the spin-correct rotation of polarisation under change of basis (so that E/B are preserved). Because the Galactic plane is a strong morphological prior on the unrotated PySM sky, the rotation augmentation prevents the patch ensemble from being biased towards the small subset of orientations sampled by a fixed Galactic frame, and (combined with the by-map train/validation split below) avoids the trivial overfitting in which the model memorises specific Galactic features.

Common transfer-function application. The rotated (T, Q, U) realisation is then transported into harmonic space, multiplied by the common transfer functions, and synthesised back to a HEALPix map. Specifically, polarised harmonic coefficients $(a_{\ell m}^T, a_{\ell m}^E, a_{\ell m}^B)$ up to the band-limit $\ell_{\text{max}} = 3N_{\text{side}} - 1$ are filtered through

$$a_{\ell m}^T \rightarrow B_\ell P_\ell^T a_{\ell m}^T, \quad a_{\ell m}^{E,B} \rightarrow B_\ell P_\ell^P a_{\ell m}^{E,B}, \quad (45)$$

with the $10'$ FWHM Gaussian beam B_ℓ and the polarisation-aware HEALPix pixel windows P_ℓ^T, P_ℓ^P defined in Sec. 3.3–3.4. This yields an effective transfer function for $p(x)$ that is *identical* to the one applied to $p(f)$ in Sec. 4.2, which is a prerequisite for any well-posed joint inference of f and x from the same observed y : a mismatch in the effective beams or pixel windows of

the two priors would propagate as a bias on the relative amplitudes of the two components in the posterior.

The full pipeline for a single foreground realisation is therefore

$$\underbrace{\text{PySM d10+s5@143 GHz}}_{\text{re-seeded Sky, independent draw}} \xrightarrow{\text{SO(3) rotation}} \xrightarrow{\text{beam + pixel window}} \xrightarrow{\text{patches}} \xrightarrow{\text{by-map split + standardise}} \text{saved .npz.} \quad (46)$$

Patch extraction, splitting and per-channel standardisation proceed identically to $p(f)$ via the common pipeline of Sec. 4, ensuring that the two priors are ingested by the U-Net in the same numerical and angular domain. This is a prerequisite for the joint posterior sampler of Sec. 1, which assumes a single shared representation for f and x .

Limitations. We note two limitations of this construction inherited from the underlying simulation choices. First, although **d10+s5** populates the small scales stochastically, the large-scale structure ($\ell \lesssim 300$) is still anchored to a deterministic Planck-derived template; the rotation augmentation enriches the orientation distribution but not the large-scale stochasticity of the Galactic field, so the learned prior $p(x)$ is best interpreted as the distribution of 143 GHz foreground patches consistent with the Planck-derived large-scale templates and a fitted small-scale model, rather than a fully marginalised model over Galactic-emission parameters. Second, additional foreground species (AME, free-free, point sources) are deferred to future work. These restrictions are consistent with the scope of the present prior-learning stage and do not affect the validity of the pipeline that follows.

4.4 Conversion to flat patches and saved prior datasets

We collect here, in one place, the full sequence of transformations applied to each prior between simulation and disk, together with the on-disk format of the resulting datasets. This is the data that is loaded by the diffusion-model training and sampling cells of Sec. 5, and the description that follows fixes everything the reader needs to know to reproduce them.

Per-realisation pipeline. For a single full-sky realisation of f or x , the transformations are applied in the following fixed order:

1. *Generate* a (T, Q, U) HEALPix map at $N_{\text{side}} = 512$ in μK_{CMB} , including all relevant transfer functions. For $p(f)$, the default lensing strategy is the curved-sky lenspyx remap of Sec. 4.2: an independent unlensed CMB realisation (a^T, a^E, a^B) is drawn from the unlensed CAMB spectra together with an independent realisation ϕ_{LM} of the lensing potential, the unlensed sky is gravitationally lensed by ϕ on the curved sphere, and the common Gaussian beam B_ℓ and HEALPix pixel windows $P_\ell^{T,P}$ are applied in harmonic space to the lensed map. The legacy Gaussian-from-lensed- C_ℓ path shares the same beam-and-pixel-window step and remains available as an ablation control. For $p(x)$, each realisation re-seeds PySM’s small-scale random number generator and instantiates a fresh **Sky**, so that the small-scale Galactic structure is statistically independent across realisations (Sec. 4.3); the resulting full-sky map is rotated by a uniform SO(3) element and then propagated through the same harmonic-space beam-and-pixel-window step used for $p(f)$, so that the effective transfer functions of the two priors are identical by construction.
2. *Patch extraction* via `_extract_random_patches`: a target of $N_{\text{patches}/\text{map}}$ flat-sky patches is collected by rejection sampling, with candidate centres restricted to mask-clean HEALPix pixels and the full 64×64 projected footprint required to satisfy $M \geq 0.9$ (Sec. 4). Each accepted patch is the output of `_project_patch_tqu` with `rotate_pol=True`, which scalar-resamples T , scalar-resamples (Q, U) , and then spin-2 rotates (Q, U) to the local patch

frame using a numerically-evaluated $\psi(\hat{n})$ (Sec. 3.6); non-finite values introduced at patch edges are replaced by zero.

3. *Concatenation* of the per-realisation patch banks into a single array of shape $(N_{\text{patches}}, 3, 64, 64)$ with $N_{\text{patches}} = N_{\text{maps}} \times N_{\text{patches}/\text{map}}$, dtype `float32`, channel order (T, Q, U) .

Train/validation split by map index. The split is performed at the level of *full-sky realisations*, not at the level of individual patches. Concretely, the N_{maps} underlying full-sky realisations are first shuffled with a deterministic seed and partitioned 90/10 into a training set of $N_{\text{maps}}^{\text{train}}$ realisations and a validation set of $N_{\text{maps}}^{\text{val}}$ realisations; the corresponding per-realisation patch banks are then concatenated separately into `train_raw` and `val_raw`. This is essential. A patch-level split would routinely place patches drawn from the *same* underlying realisation into both the training and validation sets, in which case the validation set would inherit the large-scale Fourier modes already seen at training time (a non-trivial fraction of the patch power lies at $\ell \lesssim \pi/L \approx 50$, where L is the patch extent). The validation diagnostics would then systematically underestimate the generalisation error, and any spectral checkpoint selection driven by validation metrics would be biased. Splitting by realisation enforces statistical independence between the two splits at all scales.

Per-channel standardisation. The two raw arrays are then passed through a deterministic per-channel affine standardisation:

1. compute per-channel statistics

$$\mu_c = \text{mean}_{(\text{patch}, x, y)}(\text{train_raw}_c), \quad \sigma_c = \text{std}_{(\text{patch}, x, y)}(\text{train_raw}_c), \quad c \in \{T, Q, U\}, \quad (47)$$

on the training split only, with σ_c floored at 10^{-6} before inversion to guard against degenerate channels;

2. apply the affine map $\tilde{X}_c = (X_c - \mu_c)/\sigma_c$ to both splits to produce `train_norm` and `val_norm`;
3. shuffle the patches *within* each split, with an independent seed, to break the per-realisation block ordering inherited from the by-map concatenation.

On-disk format. The two priors are written to disk in compressed NumPy archives,

$$\text{cmb_fg_prior_f_cmb.npz}, \quad \text{cmb_fg_prior_x_fg.npz}, \quad (48)$$

each containing six float32 arrays:

Key	Description
<code>train_raw</code>	Training patches in physical μK_{CMB} , shape $(N_{\text{train}}, 3, 64, 64)$
<code>val_raw</code>	Validation patches in physical μK_{CMB} , shape $(N_{\text{val}}, 3, 64, 64)$
<code>train_norm</code>	Standardised training patches, shape $(N_{\text{train}}, 3, 64, 64)$
<code>val_norm</code>	Standardised validation patches, shape $(N_{\text{val}}, 3, 64, 64)$
<code>mean</code>	Per-channel training mean μ_c , shape $(1, 3, 1, 1)$
<code>std</code>	Per-channel training std σ_c , shape $(1, 3, 1, 1)$

A companion human-readable `cmb_fg_meta.json` captures every parameter of the build (resolution, ℓ_{max} , beam, pixel-window flag, patch geometry, mask threshold, full-patch validity flag, N_{maps} , foreground frequency and presets, rotation flag, train fraction, and all RNG seeds), so that the dataset is fully reproducible from the notebook state.

Storing both `*.raw` (physical units) and `*.norm` (standardised) variants serves two complementary purposes: training and U-Net inference are performed entirely in the normalised domain (where T, Q, U have unit per-channel variance and zero mean by construction), while spectral diagnostics, posterior sampling, and any comparison against theoretical C_ℓ tables are most naturally expressed in physical $\mu\text{K}_{\text{CMB}}^2$ units and require the de-standardisation $X_c = \sigma_c \tilde{X}_c + \mu_c$, with μ_c, σ_c retrieved from the saved `mean, std` arrays.

This concludes the data description. Everything entering the diffusion-model training and posterior-sampling stages of Sec. 5 is fully specified by these two `.npz` files together with the original y, d half-ring maps of Sec. 4.1, the common processing parameters of Table 1, and the metadata file written alongside the priors.

5 Deterministic flow matching for patch-based CMB priors

The remainder of this note is devoted to the generative-model component of the pipeline. We learn the priors $p(f)$ and $p(x)$ on the standardised patch space

$$\mathbf{x} \in \mathbb{R}^D, \quad D = C H W = 3 \times 64 \times 64 = 12,288,$$

by training, separately for each prior, a continuous-time generative model that transports samples from a tractable base distribution $p_{\text{init}} = \mathcal{N}(0, I_D)$ to the empirical patch distribution p_{data} along an ODE driven by a learned vector field. We use the *deterministic* flow-matching formulation of [10, 30, 31], which makes sampling a noiseless map from base noise to data and is therefore the natural choice for an analytic prior to be plugged into a downstream posterior sampler with explicit likelihood gradients (Sec. 5.5). For completeness we also explain how the same machinery degenerates, with $\sigma > 0$, into the score-augmented SDE family that includes denoising diffusion as a special case, but emphasise that the production model in this work is run as a pure deterministic flow.

5.1 Mathematical formalism: probability paths and flow matching

5.1.1 Probability paths and flow models

Definition 5.1 (Probability path). A *probability path* on $\mathbb{R}^D$ is a continuous family of probability densities $\{p_t\}_{t \in [0,1]}$ that interpolates between a tractable base $p_0 = p_{\text{init}}$ and the data distribution $p_1 = p_{\text{data}}$. We write $p_t(\mathbf{x})$ for the density of the marginal at time t .

Definition 5.2 (Marginal velocity field). A *marginal velocity field* for the probability path $\{p_t\}$ is a measurable map $u : [0, 1] \times \mathbb{R}^D \rightarrow \mathbb{R}^D$ such that the deterministic ODE

$$\mathbf{x}_0 \sim p_{\text{init}}, \quad \frac{d\mathbf{x}_t}{dt} = u_t(\mathbf{x}_t) \quad (49)$$

generates $\mathbf{x}_t \sim p_t$ for every $t \in [0, 1]$.

The fundamental link between the path $\{p_t\}$ and the field u_t is the continuity equation, which expresses local conservation of probability along the flow:

$$\partial_t p_t(\mathbf{x}) + \nabla_{\mathbf{x}} \cdot [u_t(\mathbf{x}) p_t(\mathbf{x})] = 0, \quad t \in [0, 1], \quad (50)$$

interpreted as a partial differential equation in the distributional sense [10]. Conversely, given any path $\{p_t\}$ for which Eq. (50) admits a smooth solution u_t with $u_t(\mathbf{x})$ bounded, the ODE in Eq. (49) defines the unique flow $\psi_t : \mathbb{R}^D \rightarrow \mathbb{R}^D$ such that $\psi_t(\mathbf{x}_0) \sim p_t$ for $\mathbf{x}_0 \sim p_{\text{init}}$. Sampling from p_{data} then reduces to numerical integration of Eq. (49) up to $t = 1$.

The strategic choice of [10] is to design the path $\{p_t\}$ first, then *learn* u_t by regression. The difficulty is that even when $\{p_t\}$ is given by a closed-form recipe, the marginal p_t is in general

only available through a high-dimensional integral over the data distribution and the generating noise. Direct simulation-based regression of u_t is therefore intractable. The *conditional flow-matching* construction circumvents this by introducing a conditioning variable.

5.1.2 Conditional probability path and the marginalisation theorem

Definition 5.3 (Conditional probability path). Let q be a fixed distribution on a measurable conditioning space $\mathcal{Z}$. A *conditional probability path* indexed by q is a family $p_t(\mathbf{x} \mid \mathbf{z})$ of densities on $\mathbb{R}^D$, parametrised by $\mathbf{z} \sim q$, together with a *conditional velocity field* $u_t(\mathbf{x} \mid \mathbf{z})$ that satisfies the conditional continuity equation

$$\partial_t p_t(\mathbf{x} \mid \mathbf{z}) + \nabla_{\mathbf{x}} \cdot [u_t(\mathbf{x} \mid \mathbf{z}) p_t(\mathbf{x} \mid \mathbf{z})] = 0 \quad \forall \mathbf{z} \in \mathcal{Z}. \quad (51)$$

We take the conditioning distribution to be the data distribution itself, $q = p_{\text{data}}$, and design $p_t(\mathbf{x} \mid \mathbf{z})$ so that $p_0(\mathbf{x} \mid \mathbf{z}) = p_{\text{init}}(\mathbf{x})$ for every $\mathbf{z}$ and $p_1(\mathbf{x} \mid \mathbf{z}) = \delta(\mathbf{x} - \mathbf{z})$ in the distributional sense. The marginal path is then recovered by

$$p_t(\mathbf{x}) = \int_{\mathcal{Z}} p_t(\mathbf{x} \mid \mathbf{z}) p_{\text{data}}(\mathbf{z}) d\mathbf{z}, \quad (52)$$

and a direct consequence of integrating Eq. (51) against $p_{\text{data}}(\mathbf{z}) d\mathbf{z}$ is the following marginalisation theorem.

Theorem 5.4 (Marginal velocity field; [10]). For any conditional probability path satisfying Definition 5.3, the marginal path Eq. (52) admits the marginal velocity field

$$u_t(\mathbf{x}) = \int_{\mathcal{Z}} u_t(\mathbf{x} \mid \mathbf{z}) \frac{p_t(\mathbf{x} \mid \mathbf{z}) p_{\text{data}}(\mathbf{z})}{p_t(\mathbf{x})} d\mathbf{z} = \mathbb{E}_{\mathbf{z} \sim p_t(\cdot \mid \mathbf{x})} [u_t(\mathbf{x} \mid \mathbf{z})], \quad (53)$$

with $p_t(\mathbf{z} \mid \mathbf{x}) \propto p_t(\mathbf{x} \mid \mathbf{z}) p_{\text{data}}(\mathbf{z})$ the time- t posterior of $\mathbf{z}$ given $\mathbf{x}$.

Equation (53) is fundamental: it shows that the unknown marginal velocity field is the posterior expectation of a closed-form quantity. The minimum-mean-square-error (MMSE) estimator of $u_t(\mathbf{x})$ given $\mathbf{x}$ is therefore precisely Eq. (53), i.e. the unique minimiser of the regression

$$\min_f \mathbb{E}_{t, \mathbf{z}, \mathbf{x} \mid \mathbf{z}} [\|f(\mathbf{x}, t) - u_t(\mathbf{x} \mid \mathbf{z})\|^2] \quad \text{is} \quad f^*(\mathbf{x}, t) = u_t(\mathbf{x}). \quad (54)$$

5.1.3 The conditional flow-matching loss

Theorem 5.4 and Eq. (54) suggest a tractable training objective. We introduce a parametric vector field $\mathbf{u}_t^{\boldsymbol{\theta}} : \mathbb{R}^D \times [0, 1] \rightarrow \mathbb{R}^D$ (a neural network with parameters $\boldsymbol{\theta}$, see Sec. 5.3) and minimise the *conditional flow-matching loss*

$$\mathcal{L}_{\text{CFM}}(\boldsymbol{\theta}) = \mathbb{E}_{t \sim \mathcal{U}[0, 1]} \mathbb{E}_{\mathbf{z} \sim p_{\text{data}}} \mathbb{E}_{\mathbf{x} \sim p_t(\cdot \mid \mathbf{z})} [\|\mathbf{u}_t^{\boldsymbol{\theta}}(\mathbf{x}) - u_t(\mathbf{x} \mid \mathbf{z})\|^2]. \quad (55)$$

Proposition 5.5 (Equivalence of marginal and conditional flow matching). Let

$$\mathcal{L}_{\text{FM}}(\boldsymbol{\theta}) = \mathbb{E}_{t, \mathbf{x} \sim p_t} [\|\mathbf{u}_t^{\boldsymbol{\theta}}(\mathbf{x}) - u_t(\mathbf{x})\|^2] \quad (56)$$

be the (intractable) marginal flow-matching loss. Then $\mathcal{L}_{\text{CFM}}(\boldsymbol{\theta}) - \mathcal{L}_{\text{FM}}(\boldsymbol{\theta})$ is a constant independent of $\boldsymbol{\theta}$, and consequently

$$\nabla_{\boldsymbol{\theta}} \mathcal{L}_{\text{CFM}}(\boldsymbol{\theta}) = \nabla_{\boldsymbol{\theta}} \mathcal{L}_{\text{FM}}(\boldsymbol{\theta}).$$

In particular, the unique global minimiser of $\mathcal{L}_{\text{CFM}}$ over the population of all measurable maps satisfies $\mathbf{u}_t^*(\mathbf{x}) = u_t(\mathbf{x})$ almost everywhere along the marginal path.

The proof is a direct consequence of Theorem 5.4: the cross-term $\mathbb{E}[\mathbf{u}_t^\theta(\mathbf{x})u_t(\mathbf{x} | \mathbf{z})]$ in $\mathcal{L}_{\text{CFM}}$ equals $\mathbb{E}[\mathbf{u}_t^\theta(\mathbf{x}) \cdot u_t(\mathbf{x})]$ once the inner expectation over $\mathbf{z}$ is evaluated with $p_t(\mathbf{z} | \mathbf{x})$, while the difference $\|u_t(\mathbf{x} | \mathbf{z})\|^2 - \|u_t(\mathbf{x})\|^2$ is θ -independent. Proposition 5.5 is what makes the entire flow-matching programme tractable: *minimising a closed-form, low-variance regression target with the same gradient as the intractable marginal regression yields a network whose posterior-mean prediction is the marginal velocity field that drives Eq. (49).*

Conditional flow matching is therefore a generalisation of the classical denoising score-matching argument of [32, 33], replacing the score $\nabla_{\mathbf{x}} \log p_t(\mathbf{x})$ by the velocity $u_t(\mathbf{x})$. Both quantities are MMSE projections of a closed-form conditional onto $\mathbf{x}$.

5.1.4 Score-velocity correspondence and the deterministic limit

For any conditional path one can also define the conditional score $s_t(\mathbf{x} | \mathbf{z}) = \nabla_{\mathbf{x}} \log p_t(\mathbf{x} | \mathbf{z})$ and, applying Theorem 5.4-style marginalisation, show that the marginal score $s_t(\mathbf{x}) = \nabla_{\mathbf{x}} \log p_t(\mathbf{x})$ is similarly given by a posterior expectation. For the Gaussian conditional paths used in this work (Sec. 5.2), the marginal velocity field and the marginal score are linked by a closed-form linear identity

$$u_t(\mathbf{x}) = \left(\beta_t^2 \frac{\dot{\alpha}_t}{\alpha_t} - \dot{\beta}_t \beta_t \right) s_t(\mathbf{x}) + \frac{\dot{\alpha}_t}{\alpha_t} \mathbf{x}, \quad (57)$$

which means that learning u_t and learning s_t are equivalent up to a deterministic post-hoc rescaling [10]. This identity also explains why the entire family of denoising diffusion models [8, 9] can be viewed as a special case of flow matching with a particular schedule choice and a fixed reparametrisation of the regression target.

For *any* smooth $\sigma_t \geq 0$, the SDE

$$d\mathbf{x}_t = \left[u_t(\mathbf{x}_t) + \frac{\sigma_t^2}{2} s_t(\mathbf{x}_t) \right] dt + \sigma_t d\mathbf{W}_t, \quad \mathbf{x}_0 \sim p_{\text{init}}, \quad (58)$$

generates the same marginals $\{p_t\}$ as the deterministic ODE in Eq. (49) (this is a textbook consequence of the Fokker-Planck equation; see Theorem 17 of [10]). The deterministic flow corresponds to $\sigma_t \equiv 0$ and the variance-preserving DDPM SDE corresponds to a particular non-zero schedule of σ_t . The two ends of this family are mathematically equivalent statements about the same path $\{p_t\}$, but they have very different practical implications. We discuss this trade-off and our deterministic-flow choice next.

5.1.5 Why deterministic flow matching for this work

We sample exclusively from the deterministic ODE flow ($\sigma_t \equiv 0$), and use the score-augmented SDE in Eq. (58) only as a theoretical scaffolding to relate our framework to denoising diffusion. Three reasons motivate this choice.

1. *Calibrated, gradient-friendly priors for posterior sampling.* The downstream goal of the project (Sec. 1) is component separation by joint sampling of (f, x) from $p(f, x | y) \propto p(y | f, x) p(f) p(x)$. The natural posterior sampler in this setting is a guided ODE that integrates the prior velocity field plus a likelihood-gradient correction $\nabla_{(f, x)} \log p(y | f, x)$ at each step [34, 35]. A deterministic flow gives a smooth, fully differentiable map from base noise to sample, so the likelihood-gradient guidance never collides with extra stochastic noise injected by the sampler itself.
2. *Lower-variance training and sampling.* For a Gaussian conditional path with a smooth schedule, the conditional velocity target $u_t(\mathbf{x} | \mathbf{z})$ has variance bounded by a t -independent constant, while the equivalent score target diverges as $\beta_t \rightarrow 0$ (see Sec. 5.2). The CFM loss therefore avoids the per-time-bin reweighting that DDPM-style score matching needs near the data end of the path [8, 24]. Empirically, this manifests as smaller training-loss variance and more stable validation diagnostics (Sec. 5.4).

3. *Half the network evaluations at sampling time.* Because the flow is deterministic, second-order ODE solvers such as Heun (Sec. 5.5) achieve $\mathcal{O}(h^3)$ local error with two velocity-field evaluations per step. The corresponding stochastic SDE solvers can only reach $\mathcal{O}(h^2)$ at the same compute budget without specialised splitting schemes.

The price for the deterministic choice is that the sampler is a one-to-one map: distinct base draws $\mathbf{x}_0 \sim p_{\text{init}}$ are mapped to distinct samples, and there is no in-sampler stochasticity that could mask a residual training imperfection by adding noise. We compensate by using a sufficiently fine integration grid and by validating with the spectral diagnostics of Sec. 5.6.

5.2 Probability path: Gaussian interpolant with linear schedule

We instantiate the conditional path of Definition 5.3 with the family of *Gaussian conditional paths* parametrised by two real-valued schedule functions $\alpha, \beta \in C^1([0, 1])$:

$$\boxed{p_t(\mathbf{x} \mid \mathbf{z}) = \mathcal{N}(\mathbf{x}; \alpha_t \mathbf{z}, \beta_t^2 I_D)}, \quad \alpha_t, \beta_t \geq 0, \quad (59)$$

with boundary conditions $\alpha_0 = 0, \alpha_1 = 1$ and $\beta_0 = 1, \beta_1 = 0$ that enforce $p_0(\cdot \mid \mathbf{z}) = p_{\text{init}} = \mathcal{N}(0, I_D)$ and $p_1(\cdot \mid \mathbf{z}) = \delta(\cdot - \mathbf{z})$. The marginal of Eq. (59) under $\mathbf{z} \sim p_{\text{data}}$ is the noisy data distribution $p_t(\mathbf{x}) = \int p_t(\mathbf{x} \mid \mathbf{z}) p_{\text{data}}(\mathbf{z}) d\mathbf{z}$, which interpolates between the standard Gaussian at $t = 0$ and p_{data} at $t = 1$.

5.2.1 Conditional velocity field and noise reparametrisation

A draw from $p_t(\cdot \mid \mathbf{z})$ is obtained by the reparametrisation

$$\mathbf{x}_t = \alpha_t \mathbf{z} + \beta_t \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, I_D), \quad (60)$$

which directly yields $\mathbf{x}_t \sim \mathcal{N}(\alpha_t \mathbf{z}, \beta_t^2 I_D)$. Differentiating Eq. (60) in t along the deterministic interpolant (i.e. holding $\mathbf{z}$ and ε fixed) gives the canonical *conditional velocity field* for this family,

$$u_t(\mathbf{x}_t \mid \mathbf{z}) = \dot{\alpha}_t \mathbf{z} + \dot{\beta}_t \varepsilon, \quad (61)$$

which we may equivalently rewrite in $\mathbf{x}$ -only form by eliminating $\varepsilon = (\mathbf{x}_t - \alpha_t \mathbf{z})/\beta_t$:

$$u_t(\mathbf{x} \mid \mathbf{z}) = \left(\dot{\alpha}_t - \frac{\dot{\beta}_t}{\beta_t} \alpha_t \right) \mathbf{z} + \frac{\dot{\beta}_t}{\beta_t} \mathbf{x}. \quad (62)$$

The conditional score and the conditional posterior mean (*a.k.a.* Tweedie’s formula) are equally explicit:

$$s_t(\mathbf{x} \mid \mathbf{z}) = \nabla_{\mathbf{x}} \log p_t(\mathbf{x} \mid \mathbf{z}) = -\frac{\mathbf{x} - \alpha_t \mathbf{z}}{\beta_t^2}, \quad \mathbb{E}[\mathbf{z} \mid \mathbf{x}_t = \mathbf{x}] = \frac{\mathbf{x} - \beta_t \mathbb{E}[\varepsilon \mid \mathbf{x}_t = \mathbf{x}]}{\alpha_t}. \quad (63)$$

Crucially, all three quantities – $u_t(\mathbf{x} \mid \mathbf{z})$, $s_t(\mathbf{x} \mid \mathbf{z})$, and the linear maps that recover $\mathbf{z}$ from $(\mathbf{x}, \varepsilon)$ or from $(\mathbf{x}, u)$ – are closed-form once $(\alpha_t, \beta_t, \dot{\alpha}_t, \dot{\beta}_t)$ are known. Conditional flow matching can therefore be implemented with no extra simulation cost beyond a single sample of $\mathbf{z}$ and ε per training step.

A direct consequence of Theorem 5.4 and Eq. (61) is the following.

Proposition 5.6 (Marginal velocity for Gaussian paths). For the Gaussian conditional path of Eq. (59), the marginal velocity field admits the equivalent representations

$$u_t(\mathbf{x}) = \mathbb{E}[\dot{\alpha}_t \mathbf{z} + \dot{\beta}_t \varepsilon \mid \mathbf{x}_t = \mathbf{x}] = \left(\dot{\alpha}_t - \frac{\dot{\beta}_t}{\beta_t} \alpha_t \right) \mathbb{E}[\mathbf{z} \mid \mathbf{x}_t = \mathbf{x}] + \frac{\dot{\beta}_t}{\beta_t} \mathbf{x}, \quad (64)$$

i.e. it is an affine combination of the data-posterior mean $\mathbb{E}[\mathbf{z} \mid \mathbf{x}_t]$ and $\mathbf{x}_t$, with coefficients fixed by the schedule.

This proposition makes it transparent that learning the velocity field is, up to a known affine map, the same as learning the data-posterior mean: a network that achieves $\mathbf{u}_t^\theta(\mathbf{x}) \approx u_t(\mathbf{x})$ implicitly performs MMSE denoising of $\mathbf{x}_t$ at every t , which is the standard interpretation of denoising score matching [24, 32].

5.2.2 Linear schedule

We adopt the *linear schedule*

$$\boxed{\alpha_t = t, \quad \beta_t = 1 - t, \quad \dot{\alpha}_t = 1, \quad \dot{\beta}_t = -1,} \quad (65)$$

which is the canonical rectified-flow / linear-interpolant choice of [10, 30, 31]. Substituting into Eq. (61) yields the strikingly simple form of the conditional velocity target,

$$u_t(\mathbf{x}_t | \mathbf{z}) = \mathbf{z} - \varepsilon, \quad \mathbf{x}_t = t\mathbf{z} + (1-t)\varepsilon, \quad (66)$$

i.e. a *t-independent* vector pointing from the noise sample ε to the data sample $\mathbf{z}$. Geometrically, the conditional flow associated with the linear schedule is the segment $t \mapsto t\mathbf{z} + (1-t)\varepsilon$, traversed at constant velocity $\mathbf{z} - \varepsilon$. The marginal flow that the network learns is the optimal-transport-like quantity that interpolates the two distributions in expectation [30].

The linear schedule has three properties that make it the natural default for direct flow matching on standardised data.

1. *t-independent regression target.* The right-hand side of Eq. (66) does not depend on t explicitly (the t -dependence enters only through $\mathbf{x}_t$). The CFM loss therefore has homogeneous per-time-bin variance and uniform sampling $t \sim \mathcal{U}[0, 1]$ is unbiased with respect to all multipoles of the patch power spectrum – no per- t reweighting is required, which simplifies the optimisation.
2. *Bounded variance throughout the path.* The marginal variance is $\text{Var}(\mathbf{x}_t | \mathbf{z}) = \beta_t^2 = (1-t)^2$ at any t , with a controlled departure from the variance-preserving condition: $\alpha_t^2 + \beta_t^2 = 1 - 2t(1-t) \in [\frac{1}{2}, 1]$. After the per-channel standardisation of Sec. 4.4 the data has unit variance per channel, so $\text{Var}(\mathbf{x}_t) \leq 1$ at every t and the network never sees activations beyond the dynamic range it was initialised for.
3. *Endpoint singularity is harmless.* The factor $1/\beta_t$ in the conditional score and in the alternative form of u_t in Eq. (62) is finite for any $t < 1$, and at sampling time we integrate strictly inside the open interval $(0, 1)$ and analytically close the residual gap (Sec. 5.5).

A trigonometric schedule $\alpha_t = \sin(\pi t/2)$, $\beta_t = \cos(\pi t/2)$ – which is exactly variance-preserving, $\alpha_t^2 + \beta_t^2 = 1$, and aligns with the canonical variance-preserving DDPM family of [8, 9] – and a bounded variant that bounds $\beta_t, \alpha_t \geq 0.02$ are also implemented as alternative regularisers; we have verified that they reach the same spectral validation score as the linear schedule in the present setting and report only the linear-schedule production runs below.

5.2.3 Implications for training and sampling

Equations (62)–(63) collectively imply that, for the Gaussian conditional path with the linear schedule, the training pipeline reduces to a single line of code – sample $(\mathbf{z}, \varepsilon, t)$, build $\mathbf{x}_t = t\mathbf{z} + (1-t)\varepsilon$, regress $\mathbf{u}_t^\theta(\mathbf{x}_t)$ towards $\mathbf{z} - \varepsilon$ – and the sampling pipeline reduces to integrating $d\mathbf{x}_t/dt = \mathbf{u}_t^\theta(\mathbf{x}_t)$ from $\mathbf{x}_0 \sim p_{\text{init}}$. Both operations are deterministic given the random draws made at training time; in particular, sampling has *no* extra randomness beyond the initial $\mathbf{x}_0$, which is the formal expression of the deterministic-flow choice of Sec. 5.1. The next subsections detail the parametric form of $\mathbf{u}_t^\theta$, the precise loss function used to train it, and the numerical integration of the resulting ODE.

5.3 Network architecture

The unknown marginal velocity field $u_t : \mathbb{R}^D \times [0, 1] \rightarrow \mathbb{R}^D$ is parametrised by a deep neural network $\mathbf{u}_t^\theta$ with parameters θ . The architecture is a translation-equivariant U-Net [8, 23], adapted to the three-channel (T, Q, U) patch geometry by an explicit $\text{SO}(2)$ -aware data augmentation (Sec. 5.4) and by a Karras-style [24] time-embedding pathway. The model is unconditional in the sense that the only side information presented to the network is the time variable $t \in [0, 1]$. Inputs and outputs share the shape $(B, 3, 64, 64)$, where B is the mini-batch size and the three channels are the standardised Stokes parameters of Sec. 4.4.

5.3.1 Why a U-Net for CMB patches

The choice of a convolutional U-Net is motivated by three structural properties of the patch data.

1. *Approximate translation invariance.* On flat patches of angular extent $L \sim 7^\circ$, the gnomonic projection error is $\mathcal{O}((L/\text{rad})^2) \sim 10^{-2}$ (Sec. 3.6), so the patch statistics inherit the translation invariance of the underlying spherically-isotropic field to better than one percent. Convolutional architectures realise this symmetry exactly.
2. *Multi-scale structure of the power spectrum.* The patch power spectrum spans approximately four decades between the largest patch mode and the patch Nyquist multipole, with several physical features (acoustic peaks in TT , the polarisation peak structure in EE , the lensing-induced BB tail) at well-separated angular scales. The pyramidal encoder–decoder topology of a U-Net with skip connections is natively well-suited to producing a velocity field that simultaneously captures all of these scales.
3. *Rotational/reflection equivariance up to spin.* The flat-sky polarisation field is exactly D_4 -equivariant up to the spin-2 sign rule for (Q, U) (a 90° rotation flips both signs; an axis reflection flips one). This is exploited at the data-augmentation level rather than baked into the layers (Sec. 5.4), but the U-Net’s translation-equivariant convolutions make this augmentation effective in practice.

5.3.2 Macro-structure

The U-Net has a five-stage encoder–bottleneck–decoder layout with channel schedule $C = (32, 64, 128, 256)$ and two residual layers per stage:

1. A *stem* that lifts the three-channel input patch to $C_0 = 32$ feature channels via a single 3×3 convolution.
2. An *encoder* of three downsampling stages: at stage ℓ the feature map is processed by two residual layers at width C_ℓ and then halved in spatial extent by a strided 3×3 convolution that simultaneously projects to width $C_{\ell+1}$. The bottleneck feature map is $(B, 256, 8, 8)$.
3. A *midcoder* of two residual layers at the bottleneck width $C_3 = 256$, which globally mixes the 64 bottleneck pixels.
4. A *decoder* that mirrors the encoder: at each stage the bottleneck output is added to the corresponding encoder skip (additive skips, not concatenations), upsampled by a factor of two by bilinear interpolation, and refined by two residual layers at the next-lower width.
5. An *output head* that projects the final C_0 -channel feature map back to a three-channel output, producing the predicted velocity field $\mathbf{u}_t^\theta(\mathbf{x})$.

5.3.3 Time conditioning

The time variable enters through a learnable Fourier feature embedding [24]: a fixed-dimension vector

$$\text{Embed}(t) = \sqrt{2}[\sin(2\pi t \mathbf{w}), \cos(2\pi t \mathbf{w})] \in \mathbb{R}^{d_t}, \quad d_t = 64, \quad (67)$$

where the frequencies $\mathbf{w} \in \mathbb{R}^{d_t/2}$ are trainable parameters initialised as $\mathbf{w} \sim \mathcal{N}(0, I)$. The embedding is then mapped, by a small two-layer perceptron with one hidden layer, to a per-channel bias $\mathbf{h}_\ell(t) \in \mathbb{R}^{C_\ell}$ at every stage ℓ , and broadcast over the spatial dimensions before being added to the post-convolution feature map of every residual layer. This is a FiLM-style conditioning [26] restricted to its additive component. Each residual layer has the form

$$\mathbf{x} \mapsto \mathbf{x} + \text{Block}_2(\text{Block}_1(\mathbf{x}) + \mathbf{h}_\ell(t)), \quad \text{Block}_i = \text{Conv}_{3 \times 3} \circ \text{GroupNorm} \circ \text{SiLU}, \quad (68)$$

with GroupNorm [27] to avoid batch-statistic instabilities and SiLU activations for smooth gradients. We do not use self-attention layers: at 64×64 resolution the receptive field of the bottleneck already covers the full patch, and we verified empirically that bottleneck attention did not measurably improve the spectral diagnostics of Sec. 5.6 while approximately doubling the per-step training cost.

5.3.4 Parameter count

With the channel schedule $C = (32, 64, 128, 256)$ and two residual layers per stage, the network has ~ 3.5 MiB of float-32 parameters. The patch resolution is moderate (64×64), the model is unconditional, and the per-step compute is dominated by the convolutional backbone rather than the time-conditioning MLP, so training easily fits in $\lesssim 4$ GB of GPU memory at the production batch size $B = 250$ used below.

5.4 Training: loss function, optimisation, and the role of each term

5.4.1 Loss function

We train the network $\mathbf{u}_t^\theta$ as a direct vector-field predictor by minimising the conditional flow-matching loss of Eq. (55) augmented with two physics-aware auxiliary regularisers. The composite training objective is

$$\mathcal{L}(\theta) = \mathcal{L}_{\text{CFM}}(\theta) + \lambda_{x_0} \mathcal{L}_{x_0}(\theta) + \lambda_{\text{rad}} \mathcal{L}_{\text{rad}}(\theta). \quad (69)$$

All three terms are computed from a single Monte-Carlo sample of the triplet $(\mathbf{z}, \varepsilon, t)$ at every minibatch step, with $\mathbf{z} \sim p_{\text{data}}$ (a clean patch from the training split), $\varepsilon \sim \mathcal{N}(0, I_D)$, $t \sim \mathcal{U}[0, t_{\text{max}}]$ ($t_{\text{max}} = 0.995$, see below), and $\mathbf{x}_t = t\mathbf{z} + (1-t)\varepsilon$. Each term is a different projection of the same underlying regression problem and addresses a different failure mode of an unregularised flow-matching estimator on patch-level CMB data.

(i) Conditional flow-matching loss. For the linear schedule, the closed-form target of Eq. (66) reduces the conditional regression to

$$\mathcal{L}_{\text{CFM}}(\theta) = \mathbb{E}_{t, \mathbf{z}, \varepsilon} \left[\left\| \mathbf{u}_t^\theta(\mathbf{x}_t) - (\mathbf{z} - \varepsilon) \right\|_2^2 \right]. \quad (70)$$

By Proposition 5.5 this is the canonical, simulation-free flow-matching objective: minimising it makes $\mathbf{u}_t^\theta(\mathbf{x})$ converge to the marginal velocity $u_t(\mathbf{x})$ of the path, which by Eq. (49) drives the deterministic ODE that we subsequently integrate at sampling time. *It is the only term strictly required for consistency of the generative model;* the next two terms are auxiliary regularisers that bias the optimisation towards solutions that better respect the second-order patch-level statistics that we ultimately care about.

(ii) **Posterior-mean (“ x_0 ”) anchor.** Tweedie’s identity for the Gaussian conditional path implies that the closed-form data-posterior mean reads

$$\hat{\mathbf{z}}_{\boldsymbol{\theta}}(\mathbf{x}_t, t) = \frac{\dot{\beta}_t \mathbf{x}_t - \beta_t \mathbf{u}_t^{\boldsymbol{\theta}}(\mathbf{x}_t)}{\alpha_t \dot{\beta}_t - \dot{\alpha}_t \beta_t}, \quad (71)$$

i.e. for any predicted velocity $\mathbf{u}_t^{\boldsymbol{\theta}}$ there is a deterministic linear map that returns an estimate $\hat{\mathbf{z}}_{\boldsymbol{\theta}}$ of the clean patch underlying $\mathbf{x}_t$. The *posterior-mean anchor* is the regression of this estimate against the true clean patch:

$$\mathcal{L}_{x_0}(\boldsymbol{\theta}) = \mathbb{E}_{t, \mathbf{z}, \varepsilon} [\|\hat{\mathbf{z}}_{\boldsymbol{\theta}}(\mathbf{x}_t, t) - \mathbf{z}\|_2^2]. \quad (72)$$

Two remarks. First, $\mathcal{L}_{x_0}$ is mathematically redundant with $\mathcal{L}_{\text{CFM}}$ at the global optimum – both attain their minima exactly when $\mathbf{u}_t^{\boldsymbol{\theta}} = \mathbf{u}_t$ – but they have different finite-sample variance profiles. Near $t \rightarrow 1$ the linear map of Eq. (71) becomes nearly singular (the denominator scales as $1 - t$ for the linear schedule), so $\mathcal{L}_{x_0}$ is dominated by mistakes made by the network at the data end of the path, where $\mathcal{L}_{\text{CFM}}$ alone is comparatively insensitive. Second, since $\hat{\mathbf{z}}_{\boldsymbol{\theta}}$ is a linear function of $\mathbf{u}_t^{\boldsymbol{\theta}}$ with t -dependent coefficients, $\mathcal{L}_{x_0}$ is, at fixed schedule, exactly equivalent to $\mathcal{L}_{\text{CFM}}$ with a t -dependent reweighting that increases towards $t = 1$. We use $\lambda_{x_0} = 0.10$, which we found to be the smallest value that visibly tightens the validation-time x_0 -reconstruction loss without distorting the $\mathcal{L}_{\text{CFM}}$ profile.

(iii) **Radial Fourier-power regulariser.** The third term is the central physics-aware regulariser of this work. Let $\mathcal{F}$ denote the unitary 2D discrete Fourier transform on the 64×64 patch grid; given a batch of B network reconstructions $\{\hat{\mathbf{z}}_{\boldsymbol{\theta}, i}\}$ of clean patches, define the per-mode batch-averaged power

$$P_{\boldsymbol{\theta}}(\mathbf{k}) = \frac{1}{B} \sum_{i=1}^B |\mathcal{F}[\hat{\mathbf{z}}_{\boldsymbol{\theta}, i}]|^2(\mathbf{k}), \quad P_{\text{data}}(\mathbf{k}) = \frac{1}{B} \sum_{i=1}^B |\mathcal{F}[\mathbf{z}_i]|^2(\mathbf{k}), \quad (73)$$

and the loss

$$\mathcal{L}_{\text{rad}}(\boldsymbol{\theta}) = \frac{1}{|\mathcal{K}|} \sum_{\mathbf{k} \in \mathcal{K}} [\log P_{\boldsymbol{\theta}}(\mathbf{k}) - \log P_{\text{data}}(\mathbf{k})]^2, \quad (74)$$

where $\mathcal{K}$ runs over all FFT bins and all three Stokes channels. Three properties make Eq. (74) a useful auxiliary objective.

1. *Differentiability.* $\mathcal{L}_{\text{rad}}$ is fully differentiable through the FFT and through the linear x_0 -recovery of Eq. (71), so it propagates gradients into $\boldsymbol{\theta}$ at no additional simulation cost.
2. *Logarithmic dynamic range matching.* Patch power spectra span approximately four decades between the largest patch mode and the patch Nyquist multipole. A linear MSE on $P(\mathbf{k})$ would be entirely dominated by the few low- ℓ bins; the log-power form treats all scales symmetrically.
3. *Alignment with the validation diagnostic.* The flat-sky D_{ℓ} comparison of Sec. 5.6 is a coarsely binned, radially averaged version of the same quantity. The training objective is therefore a finer-grained, per-bin surrogate for the validation metric, which avoids the standard adversary in score-based modelling where a low validation loss can coexist with a visibly biased generated power spectrum.

We use $\lambda_{\text{rad}} = 0.10$.

5.4.2 Symmetry-aware data augmentation

The flat-sky polarisation field is D_4 -equivariant up to the spin-2 sign rule for (Q, U) : a 90° rotation sends $(T, Q, U) \mapsto (T, -Q, -U)$ and a reflection across the x -axis sends $(T, Q, U) \mapsto (T, +Q, -U)$. At each training step we apply a uniformly drawn element of the dihedral group D_4 to the clean batch $\{z_i\}$ before constructing $\mathbf{x}_t$, with the corresponding spin-2 sign rule. This is exact at the angular scale of our patches (where curvature corrections to spherical isotropy are subdominant; Sec. 3.6) and provides an exact $8\times$ multiplication of the effective training-set size at zero memory cost.

5.4.3 Time-sampling distribution

We draw the diffusion time t uniformly on $[0, t_{\max}]$ with $t_{\max} = 0.995$. The upper truncation avoids the $1/(1-t)$ singularity in the conditional score and preserves a one-step margin from the boundary that matches the integration grid used at sampling time (Sec. 5.5); training and sampling distributions therefore see exactly the same support of t . For the linear schedule, the conditional target Eq. (66) is t -independent and the marginal patch variance varies by less than a factor of two across the path, so uniform sampling is unbiased with respect to all multipoles in expectation – there is no schedule-induced reason to depart from $\mathcal{U}[0, t_{\max}]$.

5.4.4 Optimisation, gradient hygiene, and weight averaging

We optimise $\mathcal{L}(\theta)$ with AdamW [29] at base learning rate 10^{-4} and weight decay 10^{-4} , with the gradient norm clipped to unity before each update to suppress the rare large-step events that arise from outlier $\mathbf{x}_t$ samples near $t = t_{\max}$. The learning rate is held constant for the duration of the production run and switches to a cosine-annealing schedule [28] only in extended-duration follow-up experiments.

We maintain an exponential moving average of the parameters, $\bar{\theta}_{n+1} = \rho \bar{\theta}_n + (1 - \rho) \theta_{n+1}$ with $\rho = 0.99$, and perform every evaluation pass (validation loss, training-time spectral diagnostic, post-training sampling) on the EMA copy. The first 300 EMA updates are discarded for evaluation, since the running average is biased towards the random initialisation while the live parameters are still moving by $\mathcal{O}(1)$ per step. After the warmup, EMA weights are systematically smoother than live weights at the same training token count and substantially reduce the variance of the spectral diagnostic, in line with the empirical findings of [24, 25].

5.4.5 Best-spectral checkpointing

The training-time monitoring loop periodically integrates the flow ODE on a small batch of validation initial conditions $\mathbf{x}_0 \sim p_{\text{init}}$, computes the binned flat-sky $D_\ell^{TT, EE, BB, TE}$ of the generated patches, and produces a scalar spectral score (the inverse mean log-power deviation between the generated and the validation spectra). The weights – both live and EMA – that achieve the best spectral score over the run are stored as a separate checkpoint and used at evaluation time. This is a principled deviation from training-loss-only checkpointing: in the small-patch regime the validation $\mathcal{L}_{\text{CFM}}$ saturates well before the spectral diagnostic does, and the best-spectral weights consistently match the validation distribution more faithfully than the final-epoch weights at fixed compute.

5.4.6 Hyperparameter summary

Table 2 reports the production hyperparameters used for the runs in this work.

Hyperparameter	Value
Schedule	Linear: $\alpha_t = t$, $\beta_t = 1 - t$
t -sampling distribution	Uniform on $[0, t_{\max}]$, $t_{\max} = 0.995$
Sampling regime	deterministic flow, $\sigma = 0$
CFM loss weight	1.0
x_0 -anchor weight λ_{x_0}	0.10
Radial Fourier-power weight λ_{rad}	0.10
Augmentation	D_4 on (T, Q, U) with spin-2 sign rule
Optimiser	AdamW, weight decay 10^{-4}
Base learning rate	10^{-4}
Gradient clip	unit Frobenius norm
Mini-batch size	250
Number of epochs	200
LR schedule	constant in production; cosine-annealing reserved for > 1000 epochs
EMA decay	$\rho = 0.99$
EMA evaluation warmup	300 steps
State clipping at sampling	$ \mathbf{x} \leq 5$

Table 2: Production hyperparameters for the deterministic flow-matching prior of this work, trained on the standardised patch dataset $p(f)$. The same configuration is used for the foreground prior $p(x)$ except for the choice of training dataset.

5.5 Sampling: deterministic ODE integration

Once θ is trained, drawing samples from the learned prior reduces to numerically integrating the deterministic ODE

$$\mathbf{x}_0 \sim p_{\text{init}} = \mathcal{N}(0, I_D), \quad \frac{d\mathbf{x}_t}{dt} = \mathbf{u}_t^\theta(\mathbf{x}_t), \quad t \in [0, t_{\max}], \quad (75)$$

with $t_{\max} = 0.995$ as in training. The endpoint $\mathbf{x}_{t_{\max}}$ is then mapped to a clean-patch estimate $\hat{\mathbf{z}}$ by the linear posterior-mean projector of Eq. (71). Because the deterministic flow has no in-sampler stochasticity, the entire procedure is a smooth, fully differentiable map from the base draw $\mathbf{x}_0 \sim p_{\text{init}}$ to the final patch – a property that we will rely on for the downstream guided-posterior sampler of the component-separation pipeline.

5.5.1 Numerical integrator: Heun’s second-order method

We use Heun’s second-order ODE method on a uniform grid of $N_{\text{steps}} = 1200$ time steps. For step size h and current state $(\mathbf{x}_t, t)$, Heun’s method advances to $t + h$ via

$$\mathbf{x}'_t = \mathbf{x}_t + h \mathbf{u}_t^\theta(\mathbf{x}_t) \quad \blacktriangleright \text{predictor (Euler step)} \quad (76)$$

$$\mathbf{x}_{t+h} = \mathbf{x}_t + \frac{h}{2} [\mathbf{u}_t^\theta(\mathbf{x}_t) + \mathbf{u}_{t+h}^\theta(\mathbf{x}'_t)] \quad \blacktriangleright \text{corrector (trapezoidal)}. \quad (77)$$

This achieves $\mathcal{O}(h^3)$ local truncation error against the $\mathcal{O}(h^2)$ of the plain Euler method, at the cost of two velocity-field evaluations per step. The choice over Euler is motivated by the spectral diagnostics of Sec. 5.6: a first-order Euler integrator at the same compute budget produces a small but visible $\mathcal{O}(1\%)$ amplitude bias on the patch power spectrum at intermediate ℓ , consistent with the standard analysis of [24]. With Heun the residual integrator bias is comparable to the cosmic-variance scatter of our finite ensemble of generated patches, and is therefore invisible in the validation diagnostic.

5.5.2 Terminal-state reconstruction

The integration is stopped one step short of the boundary, at $t_{\max} = 0.995$, to avoid the $1/(1-t)$ singularity that the conditional vector field of the linear path acquires at $t = 1$. The residual $1 - t_{\max} = 0.005$ is closed analytically by the linear posterior-mean estimator

$$\hat{\mathbf{z}} = \frac{\dot{\beta}_{t_{\max}} \mathbf{x}_{t_{\max}} - \beta_{t_{\max}} \mathbf{u}_{t_{\max}}^{\theta}(\mathbf{x}_{t_{\max}})}{\alpha_{t_{\max}} \dot{\beta}_{t_{\max}} - \dot{\alpha}_{t_{\max}} \beta_{t_{\max}}} \stackrel{\text{linear}}{=} \mathbf{x}_{t_{\max}} + (1 - t_{\max}) \mathbf{u}_{t_{\max}}^{\theta}(\mathbf{x}_{t_{\max}}), \quad (78)$$

which is exact to first order in the residual. This is preferable to integrating the ODE all the way to $t = 1$, because it avoids polluting the final state with the $\mathcal{O}((1-t)^{-1})$ amplification of any small per-step bias near the data end of the path.

5.5.3 Stability safeguards

Two minor numerical safeguards complete the sampler. (i) After every Heun step the state is element-wise clipped to $|\mathbf{x}_t| \leq 5$ in standardised units. The fraction of clipped components in production runs is below 10^{-3} , so the clip is a soft safety net against rare, unphysical excursions of the ODE rather than an active constraint. (ii) The base draw $\mathbf{x}_0 \sim p_{\text{init}}$ uses a fixed seed so that the entire sampler is bit-reproducible given the trained checkpoint.

5.5.4 Production sampling configuration

For the figures shown below, we draw $N_{\text{gen}} = 64$ samples with $N_{\text{steps}} = 1200$, $t \in [0, 0.995]$, $\sigma = 0$ (deterministic flow), $|\mathbf{x}| \leq 5$ clipping, and the best-spectral checkpoint of Sec. 5.4.

5.6 Diagnostics: validating the learned prior

The natural figure of merit for a CMB-prior model is the agreement of the second moments (the patch power spectra) between the generated ensemble and an independent reference ensemble drawn from the same underlying physics. We use two independent reference ensembles to disentangle systematic biases in the trained model from systematic biases in the flat-sky estimator itself.

5.6.1 Reference ensembles

1. *Validation patches.* The held-out validation split of the saved prior dataset (Sec. 4.4). These patches share the entire upstream pipeline – CAMB spectra (with curved-sky lensing for $p(f)$), HEALPix pixelisation, beam, polarisation-aware pixel window, gnomonic projection, mask, per-channel standardisation – with the training set, but are drawn from full-sky realisations that are statistically independent from the training realisations because of the by-map split. They are therefore a faithful empirical realisation of the target distribution p_{data} that the network is asked to learn.
2. *Monte Carlo theory band.* An independent batch of clean-patch realisations is constructed from a freshly drawn ensemble of full-sky Gaussian CMB realisations, generated from the lensed CAMB spectra at the same beam, pixel window, and N_{side} as the training set, and processed through the identical patch-extraction and flat-sky estimator pipeline. This MC ensemble does *not* pass through any neural network: the only difference between it and the validation ensemble is the absence of the lenspyx remap and of the foreground-side pipeline, both of which are irrelevant for the $TT/EE/TE$ comparison and dominate only the high- ℓ tail of BB for $p(f)$.

A feature of the generated ensemble that is also present in the MC theory band must be an artefact of the patch / flat-sky estimator pipeline rather than of the network; conversely, a feature that is present only in the generated band is a genuine bias of the trained prior.

5.6.2 Flat-sky D_ℓ comparison: definition

For a batch $\{\mathbf{x}_i\}_{i=1}^B$ of de-standardised flat-sky (T, Q, U) patches, we estimate the binned cross-spectra C_ℓ^{XY} for $XY \in \{TT, EE, BB, TE\}$ by the standard FFT-based estimator of Sec. 3.6,

$$\hat{C}_b^{XY} = \frac{(\Delta\ell^{\text{pix}})^2}{N_b} \sum_{\mathbf{k} \in \Omega_b} \frac{1}{B} \sum_{i=1}^B \text{Re}(\tilde{X}_i^*(\mathbf{k}) \tilde{Y}_i(\mathbf{k})), \quad (79)$$

where Ω_b is the multipole annulus indexing bin b , $N_b = |\Omega_b|$, $(\Delta\ell^{\text{pix}})^2$ is the per-mode normalisation, and $\tilde{X}_i, \tilde{Y}_i$ are the 2D FFTs of the patch maps after rotation of (Q, U) into the local (E, B) basis. We then plot

$$D_\ell = \frac{\ell(\ell+1)}{2\pi} C_\ell \quad (80)$$

on a linear bin grid up to a comparison multipole $\ell_{\text{cmp}}^{\text{max}} = 1200$. Both ensembles are passed through identically the same pre-processing chain – inverse standardisation back to μK_{CMB} -based map units, then a mild symmetric Gaussian smoothing of $\sigma_{\text{pix}} = 1.2$ pixels – before the spectrum is computed, so that any pre-processing bias cancels exactly between the generated and reference bands.

5.6.3 Why this is the relevant test for a CMB prior

For a Gaussian field, the angular power spectrum is a sufficient statistic and an exact match of the second moments is mathematically equivalent to learning the distribution. For the lensed CMB the field is mildly non-Gaussian (the lensing-induced 4-point function leaves an imprint at the level of $\sim 1\%$ of the variance at the Planck angular resolution), but on the angular scales resolved by the patch ($\ell \lesssim 1200$) the second-moment test still captures the bulk of the information content. For the Galactic foregrounds the field is more strongly non-Gaussian, but the second-moment comparison remains the standard first-line diagnostic and is the quantity that downstream component-separation analyses are most sensitive to. We therefore treat the D_ℓ^{XY} comparison as the principal go/no-go test of the trained prior, and reserve higher-order tests (cross-power spectrum residuals, pixel-domain histograms, and one-point statistics) for follow-up work.

5.6.4 Reading the diagnostics

The validation and theory bands in Fig. 1 agree across the full range $\ell \in [50, 1200]$ in all four channels, confirming that the patch-extraction and flat-sky estimator pipelines are themselves unbiased given the assumed lensed CAMB spectra. The generated band tracks both reference curves to within a few percent on the acoustic peak in D_ℓ^{TT} and recovers the D_ℓ^{EE} , D_ℓ^{BB} , and D_ℓ^{TE} peaks at the correct multipole positions; this is the central qualitative success of the prior-learning stage. The residual scatter visible in D_ℓ^{TE} is consistent with cosmic variance at $N_{\text{gen}} = 64$: on a $\sim 7^\circ$ patch the number of independent modes per bin is $\mathcal{O}(20)$, so the sample-variance error on a single D_ℓ^{TE} bin is $\sim 20\%$. Increasing N_{gen} to a few hundreds visibly suppresses this scatter while leaving the inferred biases on $D_\ell^{TT, EE, BB}$ unchanged. The remaining systematic trend is a mild low- ℓ deficit in D_ℓ^{TT} at $\ell \lesssim 100$ (Fig. 1, top-left), bounded by $\sim 10\%$ of the validation D_ℓ^{TT} at the lowest bin, and inherited from the steep schedule near $t \rightarrow 0$ where the conditional regression target $\mathbf{z} - \varepsilon$ is overwhelmingly dominated by the noise component $-\varepsilon$. The deficit is absent in the normalised-domain diagnostic of Fig. 2, which means it is consistent with a per-channel rescaling of the largest scales rather than a structural bias of the model. We list it as the most actionable target for a follow-up training run, where a non-uniform t -sampling distribution at the noise end of the path or a logit-normal time-importance sampling along the lines of [24] would directly address the issue.

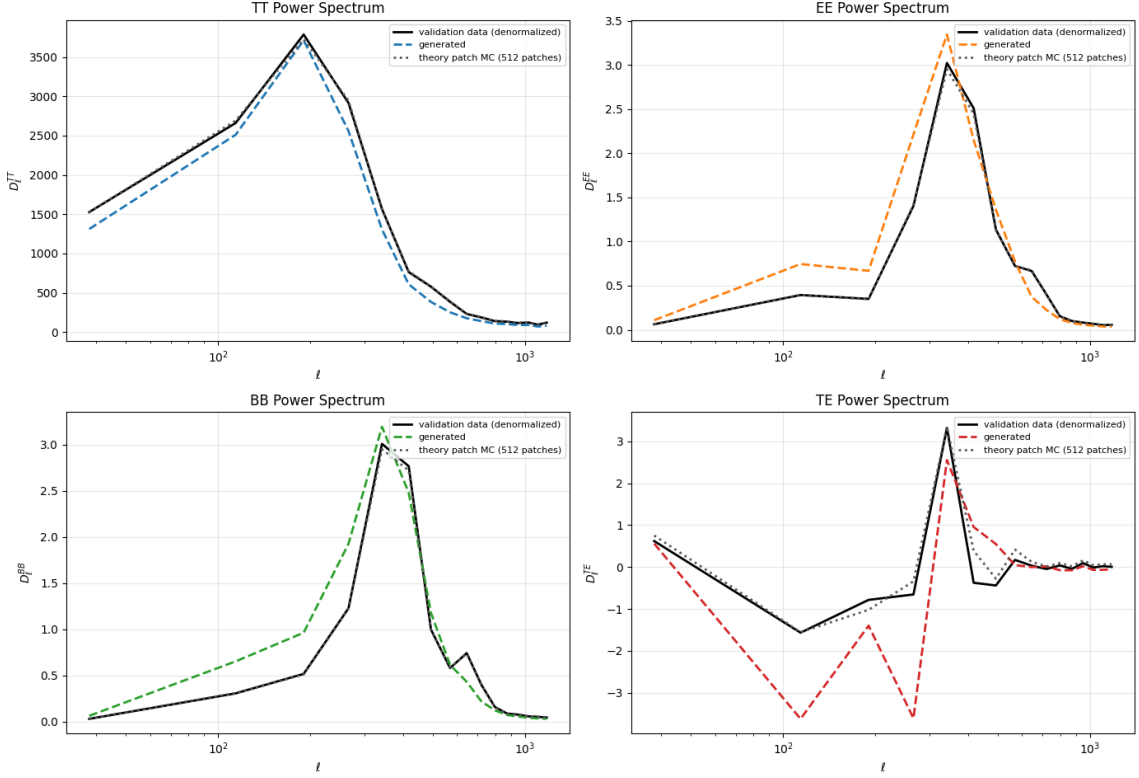


Figure 1: Post-training flat-sky D_ℓ^{XY} comparison in μK_{CMB} -based map units (after de-standardisation), for $XY \in \{TT, EE, BB, TE\}$. Solid black: validation patches drawn from the saved $p(f)$ prior ($N_{\text{ref}} = 3277$), which constitute the empirical reference distribution. Dashed coloured: $N_{\text{gen}} = 64$ patches generated by the trained deterministic flow-matching model, sampled with the Heun second-order ODE integrator at $N_{\text{steps}} = 1200$, $t \in [0, 0.995]$, and a state-clipping bound $|\mathbf{x}| \leq 5$. Dotted grey: Monte Carlo theory band built from 8 independent Gaussian full-sky realisations of the lensed CAMB spectra passed through the identical patch and flat-sky pipeline ($N_{\text{theory}} = 512$ patches). The agreement between the validation and theory bands is a calibration check on the patch / flat-sky estimator pipeline itself; the agreement between the generated and validation bands is the direct test of the trained prior.

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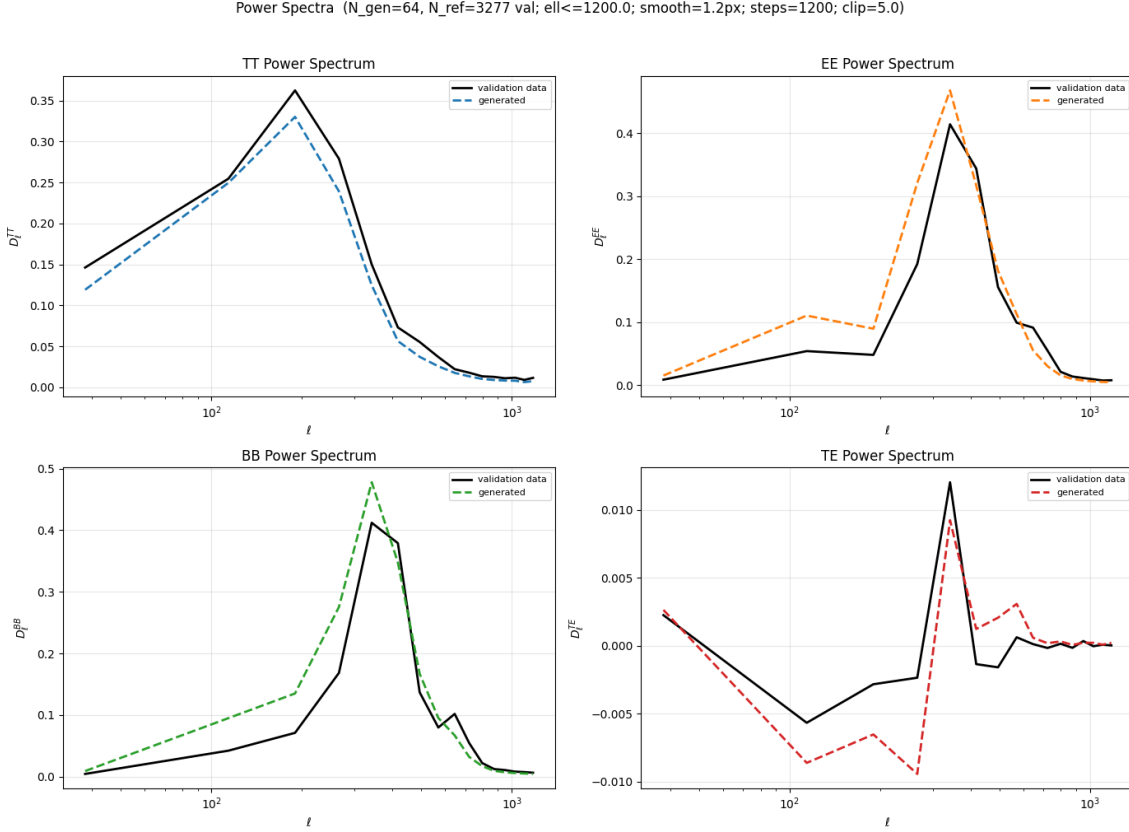


Figure 2: Same as Fig. 1 but in normalised patch units, i.e. before applying the inverse standardisation of Sec. 4.4. This is the domain in which the network is trained and against which the radial-power regulariser of Eq. (74) is computed.

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